Molecular diffusion enhanced performance evaluation of metal-organic frameworks for CO_2 capture

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ABSTRACT

Molecular diffusion is a fundamental property that limits the performance of solid sorbents in carbon dioxide capture and separation applications. Unique to each sorbent, gas diffusion is determined by the physical and chemical interactions that occur between the gas molecules and a sorbent's surface atoms. At the process level where carbon dioxide capture performance is validated, however, simulations are typically carried out using generalized parameters that omit the structure-specific, molecular kinetics occurring in each sorbent. Here, we report process-scale simulations of carbon dioxide capture performance in metal-organic frameworks (MOF) informed by molecular adsorption predictions that represent the unique structural properties of each MOF. By evaluating a total of 10,143 MOFs for post-combustion carbon dioxide capture, we demonstrate that the inclusion of the material-specific, molecular diffusion dynamics significantly alters their simulated, process-level performance. Specifically, the inclusion of molecular diffusion relegates up to 20% of the MOF candidates of the initial rank-order while the top 27 MOFs exhibit productivity and energy consumption metrics that surpass those of all known materials used in today's industrial applications. The method could be applied to evaluate a broader class of solid sorbents, including covalent organic frameworks and zeolites. For validation and reuse, we provide access to data and simulation code.

Main

Point-source capture of carbon dioxide (CO_2) is a key feature of net-zero emission strategies.[1–3] Solid sorbents, such as, for example, activated carbons, zeolites, and covalent organic frameworks have been proposed as potentially high-performing CO_2 capture materials for industrial applications.[4] Among the candidate materials, metal-organic frameworks (MOF) stand out due due to their large chemical variability enabled by their building blocks, i.e., organic linkers and metal clusters.[5] While there exist trillions of MOF descriptions[6], merely thousands of those structures have been realized and validated in the lab so far.[7] Therefore, analyzing and rank-ordering all known MOF structures with regards to their application performance remains a major challenge, even by using advanced computational methods.[8,9]

Computational screening workflows typically include calculations of geometric and adsorption properties of the specific MOF candidates to be evaluated, however, the predictions are often limited to the molecular scale.[10–14] Wilmer *et al.*[15] performed a large-scale evaluation of MOFs for CH_4 capture based solely on molecular-level adsorption uptake predictions. Similarly, Lin *et al.*[16] derived a metric from molecular-level data for assessing and rank zeolites and zeolitic imidazolate frameworks (ZIF) for CO_2 capture. While indicative of the potential at process scale, individual molecular-level metrics have yet failed to predict process-level material performance overall.[17]

Nevertheless, the application-specific gas separation performance of any given material must ultimately be assessed at the process level, by taking into consideration realistic figures-of-merits for purity, productivity, recovery and energy consumption.[18] It is, therefore, necessary to devise a multi-scale screening approach for integrating the molecular-to-process scale performance evaluation.[19]

Recent studies have provided molecular-level simulation data as inputs to macro-scale process models, however, the approach poses challenges.[20] Specifically, the CO_2 uptake of a MOF structure, simulated at molecular scale either by means of Grand Canonical Monte Carlo (GCMC) or Ideal Adsorbed Solution Theory (IAST), is used as input for pressure- or

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vacuum-swing adsorption (PSA/VSA) models at process scale that comply with the 95% CO_2-purity and 90% CO_2-recovery targets, respectively, issued by the the U.S. Department of Energy.[21–26] Burns *et al.*[24] integrated atomistic simulations with a VSA process simulator while Farmahini *et al.*[25] studied the effect of mesoscale materials properties, such as size and porosity of the material constituents, on productivity and energy efficiency of the VSA process. These approaches, however, were based on the assumption that gas adsorption in MOFs does not depend on structure-dependent differences that occur at microscopic scales. Conceptually, this was reflected by assuming a constant mass transfer coefficient for all MOF candidates under investigation.

An enhanced performance assessment at process level would require an effective mass transfer coefficient K_{eff} that captures both structure-independent, macro-pore diffusivity, as well as structure-dependent, micro-pore diffusivity.[27] Indeed, several investigations [28,29] have already revealed a dependence of diffusivity on micro-pore, molecular loading. As gas diffusion and adsorption are, ultimately, determined by a MOF's structural properties, we suggest to include micro-pore gas diffusivity, provided for each MOF structure through GCMC simulations, into the process simulation of CO_2 capture performance by means of K_{eff}, as conceptually visualized in Figure 1.

In the following, we introduce a multi-scale, computational workflow to evaluate MOFs for CO_2 post-combustion capture with temperature-swing absorption (TSA) in natural gas combined cycle (NGCC) power plants. The workflow, see Figure 2a, uses molecular-level simulations for characterizing MOFs with high CO_2 adsorption selectivity, which are shortlisted based on active learning cycles, and rank-ordered based on process simulations informed by molecular-scale, adsorption figures-of-merit. Finally, process optimizations at realistic operation conditions confirm each MOF's performance at process level. We demonstrate that inclusion of molecular kinetics improves the workflow's ability to discover false leads, which reduces computational costs significantly. Moreover, based on the TOP-20 MOF ranking results, we show for representative crystallite diameters how the inclusion of microscopic diffusivity affects process-level, CO_2 capture performance predictions.

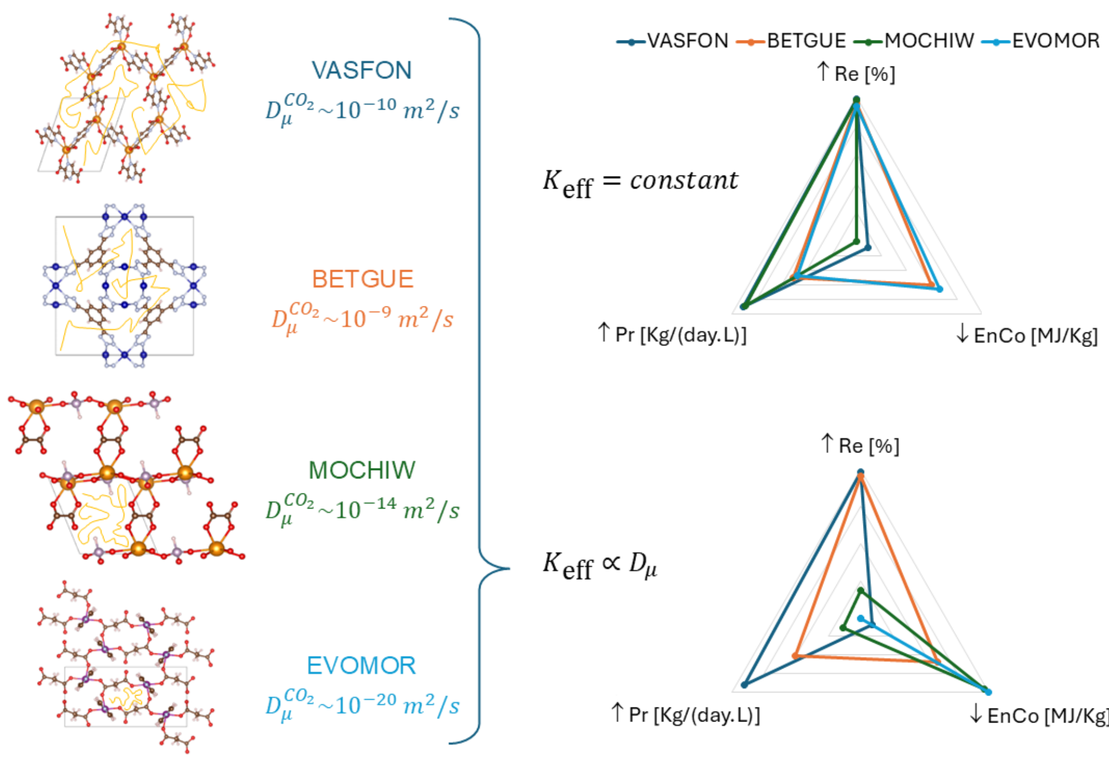

Figure 1. Impact of molecular diffusion in MOF on process performance in post-combustion CO_2 capture. Four representative MOF structures are shown on the left. The inclusion of their material-specific, micro-pore diffusivity D_μ in the effective mass transfer coefficient K_{eff}, see Equation 2, leads to significant changes in CO_2 capture outcomes at process level: **Re**covery (higher is preferable), **Pr**oductivity (higher is preferable) and **En**ergy **Co**st (lower is preferable).

Results

In the following sections, we report the results obtained by applying the computational workflow that integrates structure selection by active-learning, molecular simulation and process-level optimization. As reference database for our study, we have selected the non-disordered, all solvent removed (ASR) portion of CoRE-MOF-2019 (v1.1.2)[30,31]. From the initial dataset of 10,143 structures, we have removed 3,659 spurious MOF structures using `mofchecker`[32]. Within the remaining 6,484 MOFs, we have identified those providing the best balance between energy cost, productivity, and recovery with a purity of at least 95% as required for CO_2 capture applications at NGCC power plants.

Active-Learning based structure selection

We have built an active-learning model for predicting CO_2 and N_2 adsorption free energy. The training data was extracted by fitting a model functions to the GCMC-simulated adsorption isotherms, see Methods section. The active learning workflow uses extensive featurization combining multiple tools. In short, feature selection is performed by combining Lasso regression and variable importance discerned from initial machine-learning (ML) models. Multiple ML models are built using the selected features and free energies of adsorption, see Figure 2a. The best-performing model with the lowest uncertainty is used for Pareto-searching higher-performing materials for subsequent iterations. The material selection improves CO_2 vs. N_2 selectivity by increasing $\Delta G_{N_2} - \Delta G_{CO_2}$, as well as by selecting materials with lower prediction uncertainty. Further details are provided in the Methods section

Overall, 982 materials were subjected to GCMC simulations over 5 active learning iterations. The progression of material selectivity is shown in Figure 2b. The best performing material in each iteration is depicted by solid symbols, showing stepwise selectivity improvements. As a result of the active learning process that targets CO_2 selectivity, the distribution successively shifts towards structures with narrower pores, see Figure 2c. Most materials in iterations 4 and 5, respectively, are MOFs having a pore-limiting diameter close to 2.5 Å. We note that for structures with pore dimensions similar to the size of the gas molecules, mass transfer is controlled by micro-pore diffusion[33].

In the following, we will complement the analysis of the 982 MOF structures by simulating molecular diffusion of CO_2 and N_2, respectively, for each individual MOF structure within the refined data set. The diffusion data are then combined with the adsorption data for enabling the process-level CO_2 capture evaluation of each MOF.

Molecular- to process-level evaluation

For all selected MOF structures, we have simulated CO_2 and N_2 molecular diffusion and adsorption uptake at representative pressure and temperature levels. The self-diffusion coefficient was calculated based on mean squared molecular displacement. The heat of adsorption was extracted from the adsorption isotherms by taking the dominant site of the best fit multi-site Langmuir model which is representative of adsorption affinity. In Figure 3a, we plot the heat of adsorption for CO_2 and N_2, colored by the CO_2 diffusion coefficient at $T = 30$ °C. We focus our analysis on the lower right region with regards to the parity line which contains promising capture materials candidates due to their comparatively higher CO_2 adsorption affinity. Among those structures, MOFs with a low CO_2 heat of adsorption, between 10 and 30 kJ/mol, tend to diffuse faster, $D_\mu^{CO_2} \gtrsim 10^{-9}$ m^2/s. With increasing CO_2 heat of adsorption, the corresponding diffusion coefficient tends to decrease several orders of magnitude, due to the stronger binding affinity and/or smaller pore size. We note that the N_2 diffusion coefficient is usually 2-5 times larger, as shown in Figure S5.

The predominance of MOFs with high $\Delta H_{CO_2}/\Delta H_{N_2}$ ratio but low diffusivity ($D_\mu^{CO_2} \lesssim 10^{-11}$ m^2/s) has important implications. Generally, we expect that a high-performing CO_2 capture material should provide conditions under which gas adsorbents efficiently diffuse within the pores *and* bind strongly to the adsorption sites, even though a trade-off exists. For analyzing the impact of molecular-level diffusivity in each MOF structure at the process level, the numerical diffusivity values, D_μ, are propagated via the effective mass transfer coefficient K_{eff} in Equation 2. The K_{eff} *vs.* D_μ curves in Figure 3b exhibit two distinct regimes: micro-porosity limited transfer, and macro-porosity limited transfer. Each value of the crystallite size R_μ is associated with a characteristic threshold below which mass transfer is limited by microscopic diffusion. As a rule-of-thumb, we conclude that the larger the crystallite diameter, the stronger the impact of microscopic diffusion.

We now compare the process-level CO_2 capture productivity for two scenarios: a constant, material-independent K_{eff}, equivalent to setting $R_\mu = 0$ in Equation 2, thus neglecting microscopic diffusivity limitations, as well as a structure-dependent K_{eff} accounting for microscopic diffusivity. In both cases, we have assumed a crystallite diameter of $R_\mu = 10$ µm and a CO_2 purity of 95%. The resulting productivity values are indicated by color shades in Figure 3c. For the group of select MOFs, the plot reveals a correlation between CO_2 affinity, CO_2/N_2 selectivity, and capture productivity. Most MOFs in our selection have a low CO_2 affinity, as evidenced by a small $\Delta H_{CO_2} - \Delta H_{N_2}$ difference. Due to the high concentration of N_2 in the feed stream, their separation performance is low. Therefore, we focus our analysis on the right-hand side of the graphs. As a key result of the comparison, 14% of high-productivity MOFs, i.e., $\gtrsim 1 \frac{\text{kg}}{\text{day.L}}$, fail to sustain high productivity once molecular diffusivity is taken into account by means of a structure-dependent K_{eff}. From a computational screening perspective, we consider these structures as “false positives” that can be effectively removed from the performance-based MOF rank-order.

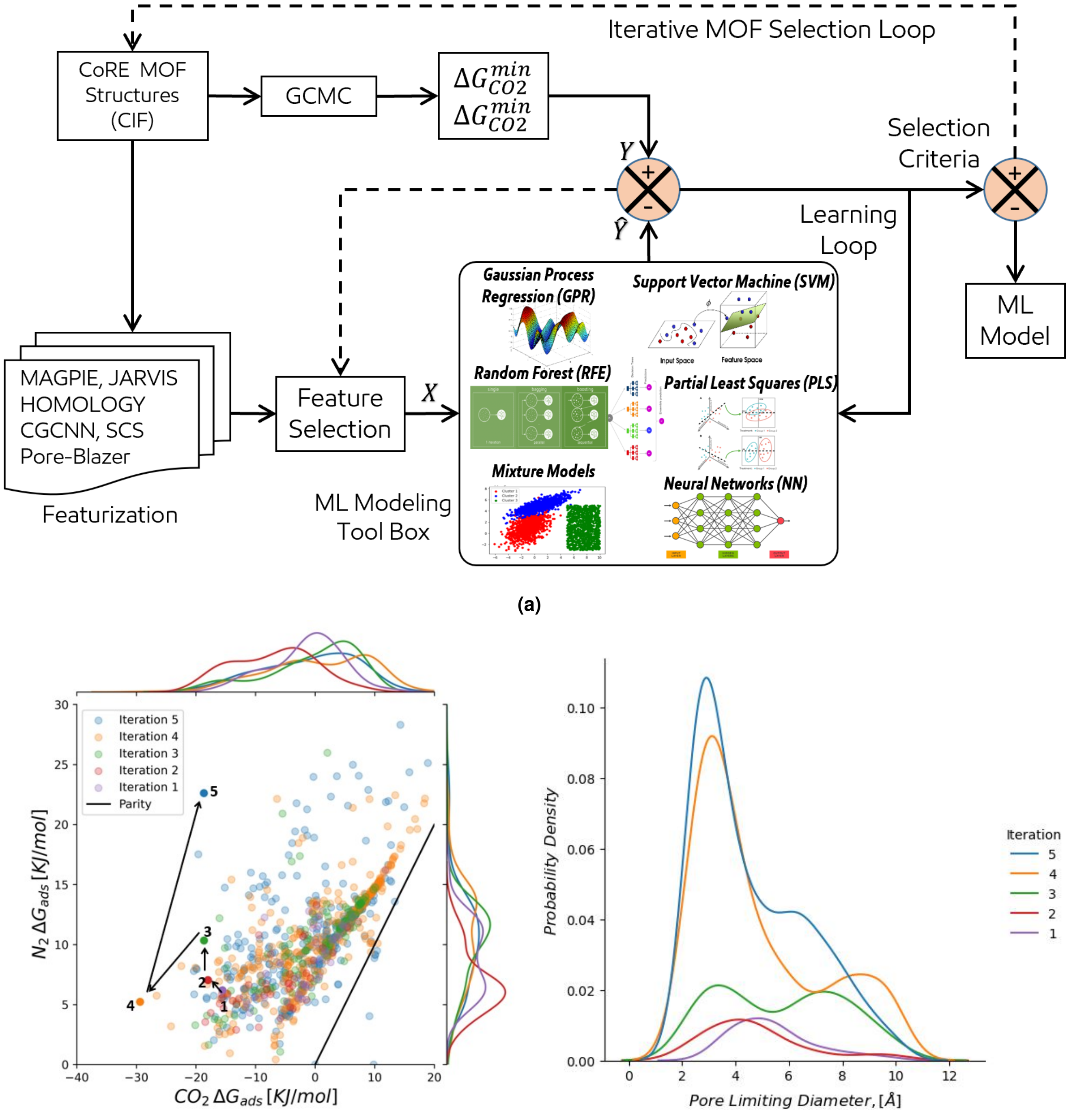

Figure 2. Active learning model for MOF selection. (a) Model architecture with iterative material selection workflow that targets high CO_2/N_2 selectivity. (b) Evolution of 5 successive model iterations and the associated MOF distributions in terms of CO_2 and N_2 adsorption free energy, respectively. The highest-ranked material in each iteration exhibits a significant increase of CO_2/N_2 selectivity with regards to prior iterations. (c) Pore-limiting diameter distribution of selected MOFs for each of the 5 active-learning iterations.

Process-level optimization and materials ranking

For a given carbon dioxide capture process, the choice of the best capture material, ultimately, depends on the trade-off between productivity and energy cost. For our select group of MOF structures, we plot in Figure 4 energy cost as function of

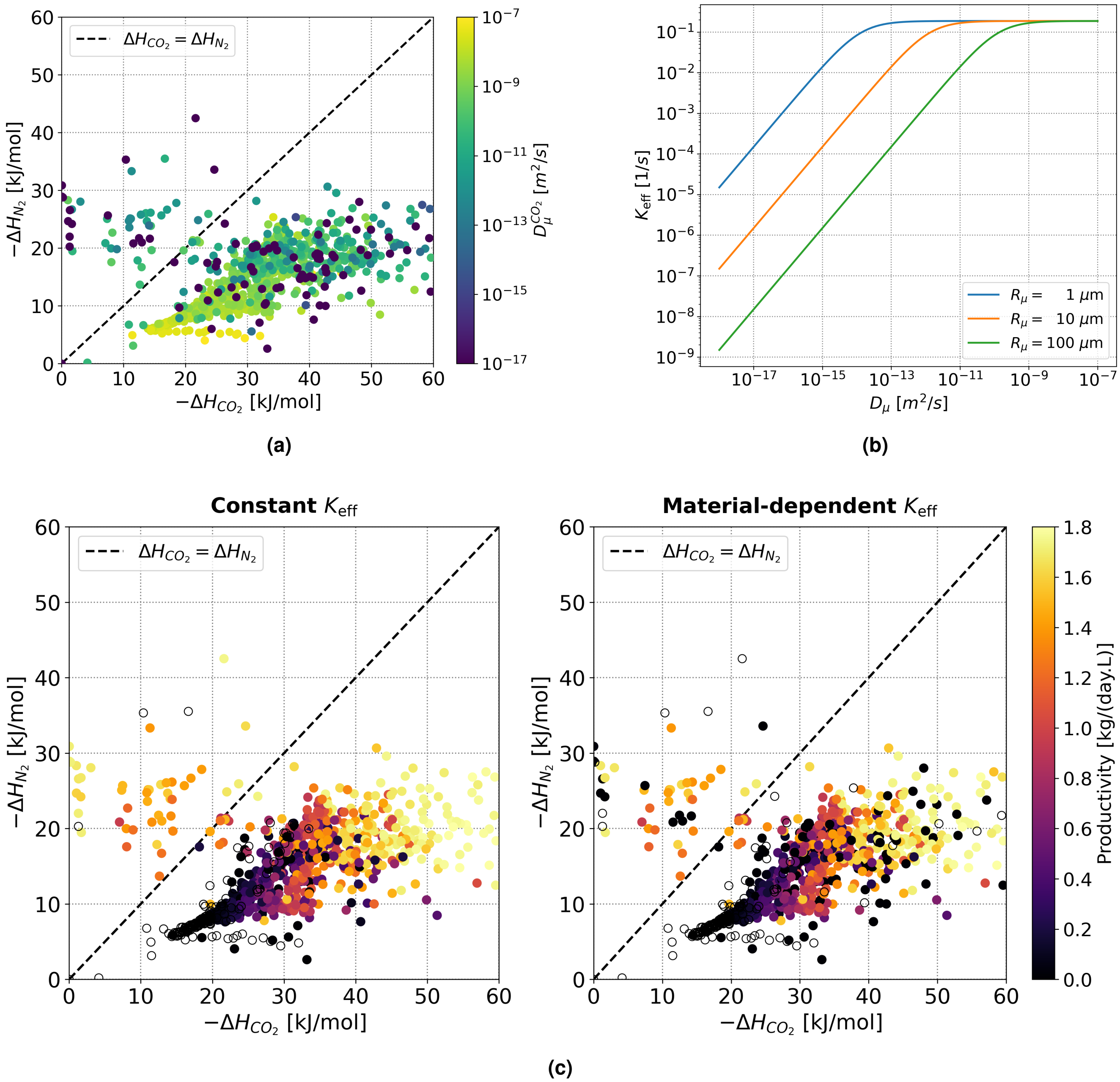

Figure 3. **Effect of molecular diffusion on CO_2 capture productivity.** (a) Distribution of heat of adsorption for CO_2 and N_2, respectively, for a set of select MOF structures. For each structure, the diffusion coefficient is indicated by color. Note, that MOFs with a high $\Delta H_{CO_2}/\Delta H_{N_2}$ ratio can have negligible microscopic CO_2 diffusion. (b) Effective mass transfer coefficient K_{eff} as function of the microscopic diffusion coefficient D_μ for three representative crystallite diameters R_μ. Below a diameter-dependent diffusivity threshold, mass transfer is limited by microscopic diffusion. (c) CO_2 capture productivity simulated at process-level with a material-independent, constant K_{eff} (left) as well as with a material-dependent K_{eff} that accounts for the microscopic diffusion coefficient of each MOF (right). Open symbols represent materials for which the process-level simulation did not yield a CO_2 purity of at least 95%.

productivity for a representative TSA process. We compare side-by-side the results obtained by assuming a constant and a material-dependent K_{eff}. We note that the distribution of energy costs spans four orders of magnitude while the productivity distribution is relatively narrow. For our in-depth analysis, we define a region of interest in the lower right corner of the plots, with CO_2 productivity higher than $1\,\frac{\mathrm{kg}}{\mathrm{day.L}}$ and energy cost lower than 5 MJ/kg of CO_2.

In Figure 4a, left panel, we plot energy cost as function of productivity assuming a constant, material-independent K_{eff}.

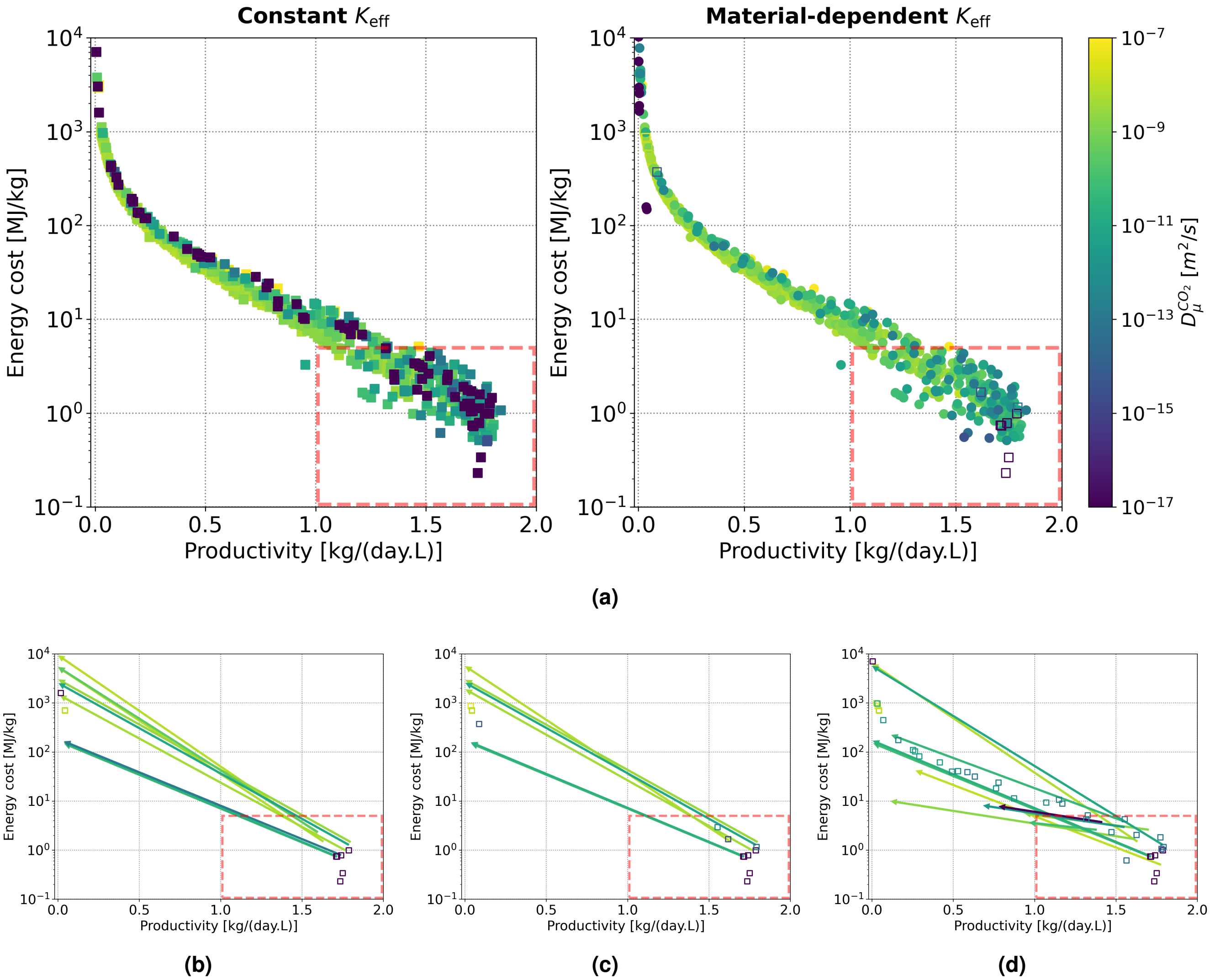

Figure 4. Effect of molecular diffusion on CO_2 capture cost and productivity. (a) Distribution of energy cost *vs.* productivity for a crystallite diameter $R_{\mu} = 10$ µm and a 95%-purity threshold assuming a constant K_{eff} (left) and a material-dependent K_{eff} (right) that accounts for microscopic diffusivity. The respective CO_2 diffusion coefficients are indicated by color. Open symbols mark the positions of excluded materials for which the process-level simulation did not yield at least 95% purity. The dashed rectangle delineates the region of interest. (b-d) Effect of microscopic diffusivity in K_{eff} on process performance for crystallite sizes 1, 10 and 100 µm, respectively. Open symbols mark the positions of excluded materials for which the process-level simulation did not yield at least 95% purity. The arrows indicate the changes undergone by MOF structures for which the inclusion of molecular diffusivity has led to significant process performance drop.

In order to guide the reader and allow for comparison, we have assigned colors for indicating the respective CO_2 diffusion coefficient of each structure, even though those diffusivity values were not used to calculate K_{eff} in this plot. We note that many of the materials in the region of interest have very small CO_2 diffusivity values. Such small diffusivity values could lead to significant changes in CO_2 capture performance if accounted for in the process model.

For analyzing the change, we have included the microscopic diffusion coefficients in the mass transfer coefficient K_{eff}, see Equation 2, and recalculated energy cost and productivity for all structures assuming a diffusivity-dependent K_{eff}. The results are shown in the right panel of Figure 4a. We note that high-performance MOFs that were initially located in the region of interest are now excluded by failing to surpass the 95%-purity threshold. In the plot on the right, their initial positions are marked by open symbols. We find that this trend depends on crystallite size, having a smaller impact for $R_{\mu} = 1$ µm, see Figure S12a, and a larger impact for $R_{\mu} = 100$ µm, see Figure S12b. In the lower panel of Figure 4, we have identified the structures whose process-level performance was most affected. We compare their performance coordinates before and after inclusion of molecular diffusivity for $R_\mu = 1,\ 10$ and 100 µm, respectively. For reference, we plot open symbols in their original positions which were taken from the left panel of Figure 4a. The arrows indicate how MOFs moved out of the high-performance regime once their structure-dependent, molecular diffusivity was included into the calculation of K_{eff}.

Overall, the percentage of MOFs that failed to yield 95% purity upon inclusion of their microscopic diffusivity increases with crystallite diameter, from 7%, to 8%, to 13%, for $R_\mu = 1,\ 10$ and 100 µm, respectively. If we consider only the region of interest, the percentages are 6%, 7% and 11%, respectively. In addition, we observe a fraction of MOFs that experience a smaller, however, still significant performance drop. They represent 5%, 5%, and 7% of the distributions initially populating the region of interest for $R_\mu = 1,\ 10$ and 100 µm, respectively. Combining both types of performance drops, the percentage of "false positive" MOF candidates that can be sorted out based on molecular diffusion amounts to 11%, 13%, and 18% of those initially located in the high-performance regime.

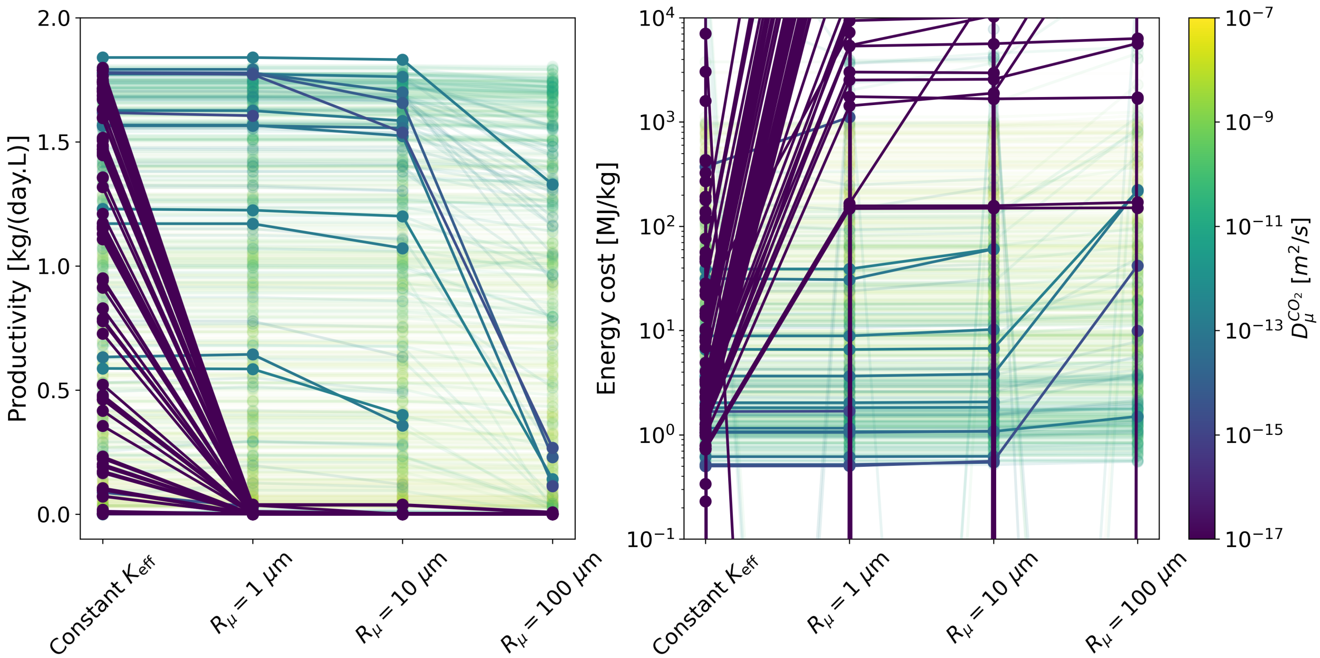

Figure 5. Effect of molecular diffusion on CO_2 capture performance ranking. Productivity (left) and energy cost (right) based on constant K_{eff}, as well as structure-dependent K_{eff} for representative crystallite sizes $R_\mu = 1,\ 10$ and 100 µm. Each curve represents a specific MOF structure and its color represents its microscopic diffusion coefficient. The higher (lower) the productivity (energy cost), the higher the performance rank of the candidate material. High diffusion coefficients are displayed with transparency for making productivity drops and energy cost increases visible in the plots. Data points for structures that fail to surpass the 95%-purity threshold are not shown.

We now discuss how the observed changes affect the CO_2 capture performance rank-order. Potentially interesting materials, with high productivity and low energy cost, may lose their high ranking upon consideration of microscopic diffusivity. Figure 5 depicts this effect whereby low diffusivity materials drop from the upper ranks upon increasing importance of microscopic diffusivity in the overall mass-transfer coefficient. Materials with higher diffusion coefficients are shown with nearly horizontal curves, as the increase in R_μ does not severely affect K_{eff} and hence process based performance. On the other hand, lower diffusivity materials are characterised by drops (rises) in productivity (energy cost) upon increasing R_μ. These results are in accordance with the behaviour of K_{eff} in Figure 3b.

As representative examples, we discuss in section S3 the properties of the two high-performing MOFs, `VASFON` and `MOCHIW`, as well as the two low-performing MOFs, `BETGUE` and `EVOMOR`, introduced in Figure 1. Specifically, we provide for each MOF the adsorption isotherms (Figure S8), diffusivities (Figure S9), breakthrough cycle profiles (Figure S10) and process-level Pareto plots (Figure S11). The plots enable a in-depth analysis of how the inclusion of molecular diffusion affects each step of the multi-scale performance evaluation.

Discussion

Using a micro- to macro-scale, computational simulation and discovery workflow for MOFs, we have demonstrated how the inclusion of structure-dependent, molecular diffusion dynamics affects post-combustion CO_2 capture figures-of-merits, i.e., purity, recovery, productivity, and energy cost, of a simulated TSA process for application at NGCC power plants.

An active-learning, structure selection model has iteratively identified a group of 982 MOFs with high molecular CO_2 / N_2 selectivity. As the identified structures exhibit narrow pore diameters, of the order of the size of the gas molecules, their CO_2 capture performance is limited by microscopic diffusion.

For quantifying the effect, we have compared the simulated TSA process performance obtained by using an effective mass transfer coefficient for all structures, as well as by calculating for each of the 982 MOFs in the selection a structure-dependent mass transfer coefficients accounting for molecular-scale diffusion dynamics. While pellet properties of representative MOFs have previously been used as decision variables for optimizing macro-pore mass transfer in vacuum-swing adsorption [25], we have focused in our work on microscopic diffusivity in crystallites, mainly due to the pore size distribution of the MOF candidates under study.

Overall, the inclusion of molecular diffusion has led to the exclusion of up to 20% of MOF candidates from the initial rank-order. This is because either the structures did not achieve the 95% CO_2 purity threshold or because of their performance loss in terms of productivity and energy cost. In the rank-order obtained by using structure-dependent diffusivity and mass transfer coefficients, 27 MOFs have achieved volumetric productivity larger than 1.5 kg/(day.L) and specific energy consumption lower than 1 MJ/kg. For comparison, reference CO_2 solid sorbents such as Zeolite 13X and the MOF materials UTSA-16 and Mg-MOF-74, have reported productivity and energy consumption of 1–3 kg/(day.L) and 0.5–1 MJ/kg, respectively, in vacuum-swing adsorption.[17] More recently, CALF-20 has achieved industrial-scale production with a productivity of 0.53 kg/(day.L) and an energy cost of 0.72 MJ/kg, respectively, in pressure-vacuum-swing adsorption.[34,35]

In Table S7, we provide our TOP-20 performance list that exemplifies the effect of microscopic diffusivity on rank-order that increases with crystallite diameter. For $R_\mu = 1,\ 10$ and 100 µm, respectively, a total of 3, 7, and 10 MOFs have dropped off the initial ranking. In case of the best-performing, TOP-1 material `BOHGOU`, the inclusion of molecular diffusivity in the process simulation has decreased volumetric productivity, from 1.84 to 1.33 kg/(day.L), and increased the energy cost, from 1.07 to 1.50 MJ/kg of CO_2. We note that in the case of $R_\mu = 100$ µm, `BOHGOU` drops out of the TOP-20 while the TOP-2 material for all other crystallite sizes, `VASFON`, moves to the TOP-1 position. This highlights the importance of MOF crystallite size on CO_2 capture performance.

We conclude that the materials at the top of the diffusion-enabled MOF ranking are high-performing CO_2 sorbents with the potential to challenge, or even surpass, the performance of the best solid sorbents deployed today. For enabling their validation and exploration, we have made our data publicly available, see **Data availability**. The computational simulation workflow for molecular diffusivity, which is based on Crystallographic Information File (CIF) format and made available as open-source code base, see **Code availability**, could be similarly applied to the evaluation of broader classes of solid sorbents, including covalent organic frameworks and zeolites.

Methods

Methodologically, we have integrated machine-learning based structure selection, molecular-level simulation, as well as process-level simulation for optimizing CO_2 capture performance. MOF properties derived from the molecular-level simulations are used as input for both the active-learning structure selection as well as the equations of the process-level temperature-swing adsorption model.

Active-learning based structure selection model

As shown in Figure 2a, active learning modeling is done by combining extensive featurization with iterative material selection. The featurization workflow combines multiple independent featurizers such as MAGPIE[36], JARVIS[37], persistent homology[38], CGCNN[39], synthesis complexity score (SCS)[40], Zeo++[41] and PoreBlazer[42]. As a fist step of the iterative materials section process, feature selection is carried out to reduce the feature space. This is done using combination of Lasso regression, variable contribution from random forest ensembles and shapely metrics. The reduced feature space is used to predict adsorption free energies ΔG for CO_2 and N_2. We employ an ensemble of models to predict the adsorption performance. Our ML modeling tool box comprised of GPR, RFE, SVM, PLS, NN with details given in section S1. The best model is selected based on performance on the test dataset, as well as the prediction uncertainty over the entire database. The selected model is used to Pareto-search and select the next set of materials to be subjected to detailed GCMC simulations. The Pareto search is performed to minimize ΔG_{CO_2} and maximize ΔG_{N_2}. Within each Pareto front the selected materials are sorted and filtered based on synthesis complexity score and the model's prediction uncertainty for adsorption free energy. We carried out 5 such active learning iterations, each including: detailed molecular simulation, (re-)building machine learning models to predict adsorption performance and material selection for subsequent iterations.

Molecular-level simulations

The molecular-level simulation framework consists of a combination of the Grand Canonical Monte Carlo (GCMC) and Molecular Dynamics (MD) methods for simulating the adsorption isotherms and diffusion coefficients, respectively. In both cases, we choose RASPA[43,44] as our simulation engine. The interaction energies between non-bonded atoms were computed using the Coulomb and Lennard-Jones (LJ) potentials with Lorentz-Berthelot[45] mixing rules

$$U_{ij}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \frac{1}{4\pi\varepsilon_0}\frac{Q_iQ_j}{r_{ij}}, \tag{1}$$

where r_{ij} is the interatomic distance, $\varepsilon_{ij} = \sqrt{\varepsilon_{ii}\varepsilon_{jj}}$ is the well depth and $\sigma_{ij} = \frac{\sigma_{ii}+\sigma_{jj}}{2}$ is the well diameter between atoms i and j. The values of LJ parameters for framework atoms are a hybrid between the Universal Force Field (UFF)[46] and DREIDING[47] (see Table S4), while the LJ parameters for the adsorbate molecules were taken from the TraPPE[48] force field (see Table S5). Electrostatic energy contributions were calculated using EQeq[49] partial atomic charges Q_i and the Ewald sum technique.

The post-processing of GCMC time series and MD trajectories was fully automated. The equilibrated averages of molecular uptake were extracted from the GCMC time series using the Marginal Standard Error Rule (MSER) as implemented in `pyMSER`[50] and shown in Figure S1. Adsorption isotherm curves were built by simulating the molecular uptake at several pressures, for each considered temperature, as shown in Figure S2. Multi-site Langmuir adsorption models (see Equation S1) were adjusted to the data points, and the best fit was kept. The resulting analytical expressions for the adsorption as a function of pressure and temperature were input to the process-level model equations.

The self-diffusion coefficients were extracted from the MD trajectories using an automated routine that locates the normal diffusion regime ($\langle r^2 \rangle \propto t$, see Figure S3) in the mean squared displacement curve, as shown in Figure S4. Then, the self-diffusion coefficients D_μ calculated at different temperatures were adjusted to an Arrhenius expression (see Equation S4), as shown in Figure S6. Finally, the fitted expression for $D_\mu(T)$ was input to the process-level models.

For more information on the approach to molecular-level modelling, please refer to section S2.

Process-level simulations

The process-level model assumed a four-step temperature-swing adsorption cycle — i.e. adsorption, purge, desorption and cooling — in a fixed bed configuration, as depicted in Figure S7c. gPROMS® Model Builder software was used for the implementation. The packed bed is modelled as one-dimensional, including heat and mass transfer. The effective mass transfer coefficient[51] is defined as

$$K_{\text{eff}} = \frac{15}{\left(\frac{R_p^2}{D_p^*} + \frac{R_\mu^2}{D_\mu}\right)}, \tag{2}$$

where the pellet radius $R_p = 2$ mm and effective pellet diffusivity $D_p^* = 5 \times 10^{-8}$ m^2/s represent the macropore (or intercrystalline) diffusion ignoring external film resistance. The crystallite radius R_μ and diffusivity D_μ represent the micropore (or intracrystalline) diffusion. Thermal management is achieved using a temperature-controlled water stream flowing through the heat exchange jacket of the bed as well as direct heat exchange through steam or cooler flue gas. The competitive adsorption of CO_2, N_2 and H_2O is modelled as a multi-component mixed Langmuir isotherm derived from the combination of pure component isotherms.

Mass balance was imposed for the gas and adsorbed phases (Equation S5). The rate of change in the concentration of the adsorbed phase was modelled as a linear restorative term proportional to K_{eff} (Equation 2) towards the equilibrium concentration (Equation S1). The energy balance at the gas and adsorbed phases, outer walls and external heat transfer fluid is described in Equation S6. The momentum balance is modelled using the Ergun equation (Equation S7) and the total pressure at each point is determined using the ideal gas law.

The desorption temperature and the four cycle step times, namely, adsorption step time, purge step time, desorption step time and cooling step time, are subject to an optimisation routine to determine the combination of values that lead to the best performance. The possible choices for these parameters contain inherent performance trade-offs, defining a Pareto front. The point with 95% purity is selected from the Pareto front as representative of the material's performance metrics.

Data availability

The data repository at https://doi.org/10.5281/zenodo.13948294 contains the MOF structures in CIF format, feature sets of the CoRE MOF database, isotherm data, diffusivity data, process performance data, cyclic simulation profiles, as well as MOF ranking results. The molecular-level simulation data, augmented with process-level metrics, is provided in JSON format.

Code availability

The molecular-level simulation framework is available at https://github.com/st4sd/nanopore-adsorption-experiment.

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Continue from here.

Additional information

Competing interests

The author(s) declare no competing interests.

Supplementary Information

S1 Active-learning structure selection model

The machine-learning approach taken here mirrors the active learning strategy used in deep-learning and other large-scale regression models. The active learning method dates to the original[52]. The basic principle is to have an intelligent user supply labels to data generated by a machine learning model. In our case the intelligent user is replaced by a combination of experts and the GCMC simulation estimating adsorption properties. The machine learning model in turn is retrained by returning the relabeled data to training the next model.

The machine learning models act as proxy leads to the molecular simulations to capture nascent structure-property indicators, by exploiting available complimentary knowledge from experts or from other datasets. They learn more and more of the physical and chemical interactions as the simulations expand. They are built iteratively, each iteration learning both new features and new epistemic components of the models. Multiple modeling paradigms are considered at every iteration, with featurization, model construction and selection repeated every time. The winning model is chosen based on accuracy on extrapolation, structural parsimony, and prediction uncertainty. The chosen model is used to re-estimate the entire space of structures under consideration, for the properties of interest and rank a new set of candidates most promising for the next set of simulations.

These objectives are embedded in the following steps of the iterative active-learning approach. All the steps below were implemented in MATLAB®[53] using its Statistics and Machine Learning[54], Signal Processing[55] and Optimization toolboxes[56].

1. Featurization: Generate a large set of features across different structural motifs that are likely to influence the adsorption characteristics of the MOF. A more detailed note on featurization is provided in the next sub-section.

2. Feature Selection: Using the results from the latest GCMC iteration, select a subset of features that are strong predictors of the two key properties, namely the free energies of CO_2 ($\Delta\ G_{CO_2}$) and N_2 ($\Delta\ G_{N_2}$) adsorption. Features are selected before the model estimation process. To enforce uniformity, features are sub-selected using a combination of Lasso[57], variable contribution from random-forest ensembles[58] and shapley[59] inspired metrics.

3. Model Estimation: The following models are constructed at every step, each with their own subset of features.

   - Support Vector Machine (SVM), linear and with gaussian kernels
   - Partial Least Squares (PLS) Regression
   - Random Forest Ensembles (RFE MATLAB's XGBoost)
   - Gaussian Process Regression (GPR)

4. Model Selection: Every model is trained on 80% of the data and testing on 20% of the data. The training step includes feature estimation and validation to extract the best set of features in the model. The best model among all regressors is selected based on performance on the testing set and estimated uncertainty. Uncertainty estimates are provided for each model by their training algorithms[54].

5. Variable Contribution: To provide early indication and insight into the nature of structural influence on the predicted properties, variable/feature contributions to the model are estimated for the selected model. The feature contribution ranking is determined using the method developed by Lundberg and Lee[59], where subsets of features enter into an n-person game[60] of training and prediction over split datasets. This method is model agnostic and is extracted over batches of data as opposed to other interpretable metrics that deal primarily with specific instances of model predictions.

6. Prediction: The selected (winning) models from (d) above is used to estimate/predict the properties of interest ΔG_{CO_2} and ΔG_{N_2}. In addition the synthetic complexity score, SCS[40] of the organics linker of MOF structure is added to the target set as a desired output. The uncertainty of the predictions is also added in from the model predictions for every structure in the database. These together form the automated design criteria for selection.

7. Pareto-search and MOF selection: Using the above design criteria, a multi-objective search is conducted over the entire space to tackle the following criteria to simultaneously

   (a) Minimize ΔG_{CO_2} - stronger adsorption

(b) Maximize ΔG_{N_2} - weaker adsorption

(c) Minimize SCS - simpler organic linker

(d) Minimize uncertainty

This is enforced by doing a formal pareto-search only on the first two criteria above. The pareto approach follows the formalism established by the Pareto Differential Algorithm (PDE)[61]. For each solution in the database the two key properties (free energies of adsorption) are calculated using the selected models as well as their attendant uncertainty estimates. The pareto front is estimated using the dual maximization of -ΔG_{CO2} and ΔG_{N2}. Each front consists of solutions that are non-dominated i.e. no solution is poorer in both criteria along the front. These solutions are "locally-efficient" as defined in the PDE[61] algorithm. each solution is locally non-dominated within a short-radius in the feature space defined by the two properties in this case. We deviate from a full pareto approach at this stage by deferring to this sequence once the pareto fronts have been identified. The entire space is filtered by deselecting candidates with very large uncertainties (>20-30% typically). Finally each front is re-ranked in the order of increasing complexity as estimated by their SCS scores.

The top-n candidates selected via the pareto-search step above form the proposed set from the multi-objective active learner. One can choose from different fronts to progressively create different trade-offs. We could also resort the fronts in terms of an average SCS score or relax uncertainty cut-offs to promote a diverse set of solutions. These criteria are considered by the experts to rank and propose new targets along with complimentary knowledge about the materials that are not codified in the models. The resulting rich set then becomes the focus of attention of the next set of GCMC simulations for following iterations.

In cultivating newer and successively better models via lower uncertainty in each iteration the active learner seeks improving accuracy. The pareto-fronts provide a balanced trade-off of "better" solutions based on properties that directly influence performance. Between these two the active learning approach iteratively walks the trade-off between exploitation and exploration, dynamically altering the search space mappings and model structure to accomplish this balance.

S1.1 Featurizing the learning space

The featurization step is performed by calculating and pooling together descriptors from different material metrics. While pooling a comprehensive set of descriptors for featurization is a one-time process performed at the beginning of the active learning cycle, feaure selection from that comprehensive set is performed at every step and sometimes in consort with model construction. For this work the following measures were considered.

1. MAGPIE[36]: These are atomic (typically distance and nearness) averaged features covered in this work[36]. We have used the MATMINER[62]'s implementation of Magpie to generate the features, calling specific featurization vectors pertinent to geometric and electronic properties. The original work[36] is generic to both crystalline and amorphous materials and quite applicable to MOFs. There are 132 features calculated though MAGPIE with each of the feature name starting with "MagpieData".

2. Force-field features: Force-fields (electronic, magnetic) represent the different activation gradients underlying many material phenomena. When featurized they form powerful embodiment of structural influences on inter-atomic interactions by condensing them into functional attributes over performance properties. The These force-fields are best obtained by molecular-level simulations under various perturnations. Joint Automated Repository for Various Integrated Simulations (JARVIS)[63], available from NIST, constructs several force-fields of interest either via molecular or atomistic simulations or accurate ML surrogate models. These features have been widely applied in correlating solid structure to semi-conductor performance metrics. However, they also generate features useful to characterize atomic, electronic, topological, mechanical, magnetic and defect-inducing properties[37]. As in several of the MAGPIE calculations, many of them also cover interactions between neighbors, expanding by atomic distance/radii to first, second and beyond neighborhood interactions. There are 1,557 features (starting with "jml") per material from this sub workflow. They comprise of 378 radial charge-distribution based, 100 radial distribution of positions, 179 angle-distribution up to first and second neighboring based each, 179 dihedral angle and 100 nearest neighbor (NN) descriptors. The remaining are 438 average chemical based including various properties and 4 simulation-box-size based to add all to 1557.

3. CGCNN[39]: Crystal Graph Convolutional Neural Networks (CGCNN) are a special class of graph neural networks, that are constructed specifically for crystalline structures with periodicity. The message passing connectivity in CGCNN mirror atom-atom connectivity and nearness. The base transfer function and architecture are that of a convolutional

Feature Type	Number of Features
Homology BarCodes	2800
JARVIS	1557
CGCNN	287
MAGPEI	132
Betti	35
Rest	179
Total	**4990**

Table S1. Feature types and count from featurization workflow

neural network. In the CGCNN however the final layer of the neural network represents the latent dimensions of the structural features. Since the CGCNN is a supervised network, these latent dimensions are obtained in training with both input and output labels, the outputs corresponding to any fundamental material properties of interest. The latent dimensions therefore represent input-output correlations corresponding to each property it is trained on.

In this work the original CGCNN architecture as presented in[39] was used, but the models were retrained on a more appropriate QMOF database[64] dataset curated further as presented in the main body of this paper. We modified the original CGCNN architecture to simultaneously train on multiple properties. This enhancement is motivated by the success of so-called transfer-learning in other areas[65,66] where designing or inversion of property to structure was also important. This also reduces the feature set dimensionality. However the testing performance with the simultaneously learnt features was much poorer than the individually trained ones, for all the architectures examined. Hence the individually trained feature dimensions (i.e. the last layer of the QMOF trained CGCNNs for the two properties, namely, band gap and total energy of the structure) were retained at the end of the featurization process to go into the larger pool. It is likely that the structural representation of small molecules as is the domain of earlier works[65,66] is of a sufficiently lower dimension that is strongly cross-correlated across multiple properties, and therefore those networks benefit from simultaneous training approaches.

Hyperparameter optimization was performed during the training using validation sets culled from the QMOF dataset. The CGCNN architecture was varied with 8, 16, 32 latent dimensions in the last layer and trained on the two property sets. The 32-dimensional latent space was chosen as providing the optimal training to testing performance across the QMOF database in both cases. The CGCNN feature set therefore consists of 292 features, nine sets of 32 features each set representing band gap, total energy, shear modulus, bulk modulus, Poisson ratio, Fermi energy, formation energy and energy per atom influencers in structure respectively. On top there are additional features in terms of final predicted total energy, band gap for the material.

4. Homology: Persistent homology[38] captures deep topological effects inherent to the pore structure of the material. It incorporates the ability of internal spaces within materials to expand or contract to occlude or accommodate other molecules in transport. They are therefore an alternative pseudo-dynamic way of capturing shape effects inherent to crystalline structures. We embed these shape effects into atom-specific features following the procedure outlined in[67] for atom-specific persistent homology (ASPH). These features potentially reveal two-body and multi-body interactions in the homology. They are highly accurate predictors of formation energy[67]. Every feature we use is calculated as a persistence barcode from a point-clouds of atomic neighborhoods. For more details see the Methods section of[67]. There are 2,800 barcode features corresponding to 90 elements each with 35 barcodes related to minimum, maximum, mean, standard deviation and sum (x5) corresponding to 7 barcodes, namely, Betti-0, Betti-1 (start, stop and length) and Betti-2 (start, stop, length). The features generated from steps 1 through 4 are pooled and id-ed by their unique structural ID, to form the base feature pool for the rest of the active learning process. In that sense, the active learner is continuously forming nonlinear maps over subsets of features at every iteration based on new adsorption data. Every such new map informs the material selection for the next iteration

In additional to above described features, there are additional 32 features related to electronic structure of the elements in the material, atomic volume, atomic wight, density, melting point etc. Each of these have min, max, mean, standard deviation and sum listed as separate features. In total the other features are 179.

S1.2 Feature importance by Shapely Explanation

The models for CO_2 and N_2 adsorption free energy were analyzed for feature importance using Shapley explanation. The Shapley values are given in Table S2 and Table S3 in the decreasing order of the important features. As it can be seen that the

Feature Name	Feature Type	Shapely Importance
LCD	Geometric	1654.6
MagpieDataAvg_dev	Electronic	1062.4
GSmagmom_sum	Electronic	1035.3
GSmagmom_mean	Electronic	1007.2
AtomicVolume_min	Electronic	793.5
AV_cm3_g	Geometric	688.6
totalenergy_latent_0	Energy	650.9
DipolePolarizability_std	Electronic	608.3
AV_VF	Electronic	603.1
jml_nn_15	Bond	534.2
jml_e3	Electronic	517.3
jml_nn_49	Bond	498.8
Column_max	Electronic	401.3
ASA_m2_g	Geometric	393.1

Table S2. CO_2 Feature importance by Shapley explanation

Feature Name	Feature Type	Shapely Importance
ASA_m2_cm3	Geometric	1257.0
LCD	Geometric	968.2
NASA_m2_cm3	Geometric	770.0
jml_nn_70	Bond	355.4
jml_C_0	Energy	313.8
totalenergy_latent_2	Energy	306.8
jml_C_1	Energy	300.8
jml_nn_48	Bond	296.7

Table S3. N_2 Feature importance by Shapley explanation

CO_2 adsorption free energy model is dominated by both gometric and electronic features while for N_2 adsorption free energy model dominated by mainly geometric features.

S2 Molecular-level simulations

The molecular-level simulations were performed with `RASPA-v2.0.47`.[43,44] The partial atomic charges were calculated using the EQeq method[49] as implemented in `pyeqeq-v0.0.9`. We used a hybrid[11] between UFF and DREIDING force fields for the framework atoms (see Table S4) and the TraPPE force field for the molecules (see Table S5).

All atoms in the framework were held fixed at their crystallographic positions. The number of unit cells used was different for each structure to ensure that the perpendicular lengths of the supercell were greater than twice the cutoff used. The cutoff for Lennard-Jones and charge-charge short-range interactions was $R_c = 12.8$ Å and the Ewald sum technique was applied to compute the long-range electrostatic interactions on a 0.1 Å grid with a relative precision of 10^{-6}. The Lennard-Jones potential was shifted to zero at the cutoff.

S2.1 Grand Canonical Monte Carlo

For each temperature and pressure value, we performed 10,000 RASPA cycles. Each RASPA cycle corresponds to $\max(20, N_{\text{ads}})$ Monte Carlo steps, where N_{ads} is the number of adsorbed molecules in the simulation box at each point of the simulation. This way, each adsorbed molecule is subject to one move attempt per cycle (on average).

In Figure S1, we show how the resulting time series for the molecular uptake as a function of Grand Canonical Monte Carlo steps is processed by an equilibration routine using the Marginal Standard Error Rule (MSER)[68] as implemented in the pyMSER[50] open-source package. For each simulation, at a given temperature and pressure, an equilibrated average loading value is calculated alongside a representative uncertainty metric. We chose the uncorrelated standard deviation (uSD) as the representative uncertainty metric. We calculate it by taking the equilibrated part of the time series, as provided by the MSER method, and splitting it into chunks of length equal to the autocorrelation time t_{corr}. Then we take the average of each (uncorrelated) chunk and calculate the standard deviation of the collection of chunk averages.

Atom	ε (K)	σ (Å)	Atom	ε (K)	σ (Å)	Atom	ε (K)	σ (Å)	Atom	ε (K)	σ (Å)
B_	47.804	3.582	**Se_**	216.376	3.591	**Ag_**	18.115	2.805	**Sm_**	4.026	3.137
C_	47.854	3.474	**Br_**	186.184	3.520	**Cd_**	114.73	2.538	**Eu_**	4.026	3.112
H_	7.649	2.847	**Ca_**	25.160	3.094	**Tl_**	342.176	3.873	**Gd_**	4.529	3.001
N_	38.948	3.263	**Sc_**	9.561	2.936	**Pb_**	333.622	3.829	**Tb_**	3.522	3.075
O_	48.156	3.034	**Ti_**	8.554	2.829	**Bi_**	260.658	3.894	**Dy_**	3.522	3.055
P_	161.024	3.698	**Cr_**	7.548	2.694	**Po_**	163.54	4.196	**Ho_**	3.522	3.038
S_	173.101	3.591	**Mn_**	6.542	2.638	**At_**	142.909	4.233	**Er_**	3.522	3.022
F_	36.482	3.094	**Fe_**	27.676	4.045	**Rn_**	124.794	4.246	**Tm_**	3.019	3.006
I_	256.632	3.698	**Co_**	7.045	2.559	**Cs_**	22.644	4.025	**Yb_**	114.73	2.990
K_	17.612	3.397	**Ni_**	7.548	2.525	**Ba_**	183.165	3.300	**Lu_**	20.631	3.243
V_	8.051	2.801	**Cu_**	2.516	3.114	**Hf_**	36.23	2.799	**Ac_**	16.606	3.099
W_	33.714	2.735	**Zn_**	27.676	4.045	**Ta_**	40.759	2.825	**Th_**	13.083	3.026
Y_	36.23	2.981	**In_**	276.760	4.090	**Re_**	33.211	2.632	**Pa_**	11.07	3.051
U_	11.07	3.025	**Sn_**	276.760	3.983	**Os_**	18.618	2.780	**Np_**	9.561	3.051
Li_	12.58	2.184	**Sb_**	276.760	3.876	**Ir_**	36.734	2.531	**Pu_**	8.051	3.051
Be_	42.772	2.446	**Te_**	286.824	3.769	**Pt_**	40.256	2.454	**Am_**	7.045	3.013
Al_	155.992	3.912	**Rb_**	20.128	3.666	**Au_**	19.625	2.934	**Cm_**	6.542	2.964
T_	155.992	3.805	**Sr_**	118.252	3.244	**Hg_**	193.732	2.410	**Bk_**	6.542	2.975
Si_	155.992	3.805	**Zr_**	34.721	2.784	**Fr_**	25.16	4.366	**Cf_**	6.542	2.952
Cl_	142.557	3.520	**Nb_**	29.689	2.820	**Ra_**	203.293	3.276	**Es_**	6.038	2.940
Na_	251.600	2.801	**Mo_**	28.179	2.720	**La_**	8.554	3.138	**Fm_**	6.038	2.928
Mg_	55.855	2.692	**Tc_**	24.154	2.671	**Ce_**	6.542	3.169	**Md_**	5.535	2.917
Ga_	201.280	3.912	**Ru_**	28.179	2.640	**Pr_**	5.032	3.213	**No_**	5.535	2.894
Ge_	201.280	3.805	**Rh_**	26.67	2.610	**Nd_**	5.032	3.186			
As_	206.312	3.698	**Pd_**	24.154	2.583	**Pm_**	4.529	3.161			

Table S4. Lennard-Jones parameters describing well depths (ε) and diameters (σ) for framework atoms.

Atom	ε (K)	σ (Å)	q (e)
C_co2	27.000	2.800	0.700
O_co2	79.000	3.050	−0.350
N_n2	36.000	3.310	−0.482
N_com	——	——	0.964

Table S5. Lennard-Jones parameters describing well depths ε, well diameters σ and point charges q for molecule atoms.

We simulated single-component gas adsorption isotherms for CO_2 and N_2 at $T = 298,\ 323,\ 348$ and 373 K, and from $P = 0.01$ to 50 bar at logarithmically spaced intervals. The simulations at different pressures were chained, so that the final state of the simulation at pressure P_n was used as the initial state of the simulation at pressure P_{n+1}. This procedure allowed reaching equilibrium faster than if we started from an empty simulation box with zero molecules at each new pressure value.

The simulated adsorption isotherms were fit to Langmuir models[69] with $S \in \{1,2,3\}$ sites

$$q(P,T) = \sum_{j=1}^{S} q_j^{\text{sat}} \frac{K_j(T) \cdot P}{1 + K_j(T) \cdot P}, \tag{S1}$$

where q_j^{sat} and $K_j(T)$ are the saturation capacity and equilibrium parameter associated with site j, respectively. The S-site Langmuir model that achieved the lowest RMSE was selected to represent the simulated isotherms, as shown in Figure S2. The equilibrium parameters $K_j(T)$ can be modelled interchangeably by the Arrhenius[70] and Van't Hoff[71] equations by expressing the pre-exponential factor of the former as the temperature-independent term of the latter, as in $b = \exp(\Delta S/R)$. The resulting analytical expressions for the adsorption as a function of pressure and temperature were input to the process-level model equations.

S2.2 Molecular Dynamics

The Molecular Dynamics simulations for the calculation of diffusion coefficients were performed aiming for a constant (and finite) volumetric loading of adsorbed molecules: $1/R_c^3$. Given that $R_c = 12.8$ Å and the simulation box has at least $(2R_c)^3$

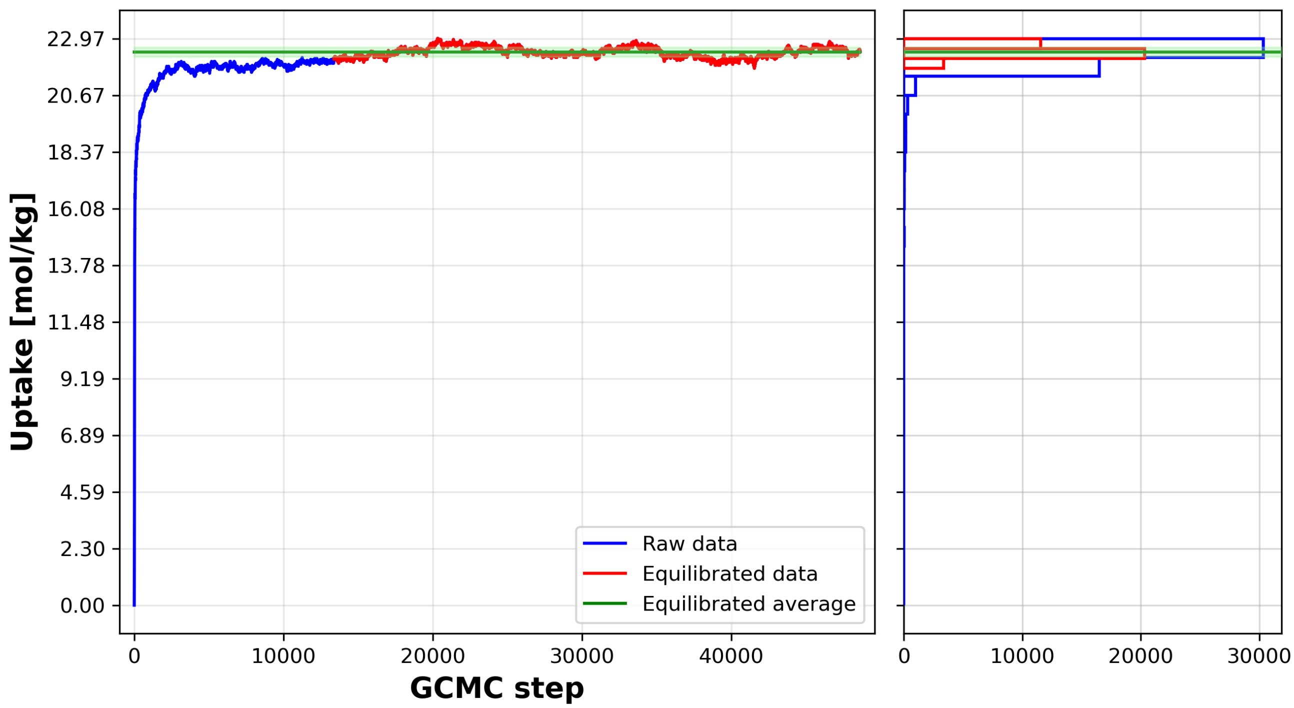


Figure S1. Time series equilibration. The molecular uptake as a function of GCMC steps is treated as a time series (blue) and processed using the MSER rule. After the truncation point is determined, the equilibrated part of the time series (red) is averaged to obtain the representative thermodynamics observable (green).

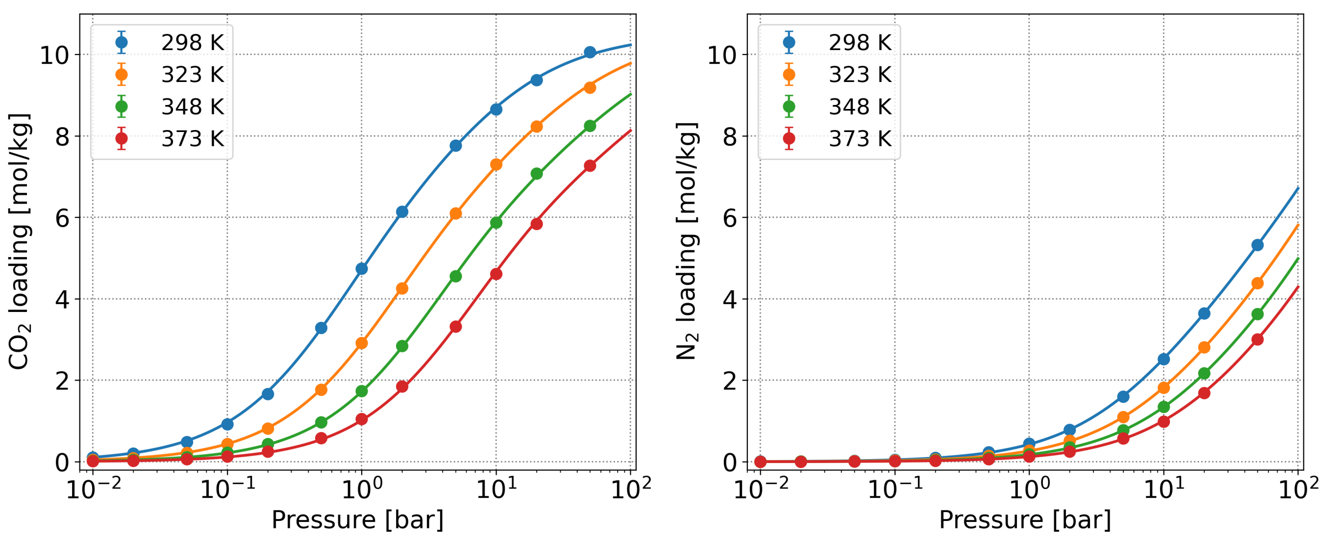


Figure S2. Simulated isotherms and adjusted Langmuir models. In this example for material "FEWWOV", symbols represent the simulated adsorption values along an isotherm and lines represent the best fit Langmuir isotherm model, at different temperatures, for CO_2 (left) and N_2 (right).

volume, this leads to a volumetric loading of at most 792 mol/m^3, which translates to a gravimetric loading of the order of 1 mol/kg. This led to a different number of molecules per simulation box for each material, given that each structure had a different unit cell size. The NVT ensemble was used with a Nosé–Hoover thermostat, and the integration time step was 10 fs. We performed 1000 GCMC cycles for initialization, 9000 MD steps (90 ps) for equilibration and 5,000,000 MD steps (50 ns) for production. The mean squared displacement (MSD) was calculated based on the trajectories of the molecules by applying the n-order algorithm.[72]

S2.2.1 Diffusion coefficient calculation

The microscopic diffusion coefficient can be computed using the Einstein equation[73]

$$D_\mu = \frac{1}{2d} \lim_{t\to\infty} \frac{\mathrm{d}}{\mathrm{d}t} \left\langle \frac{1}{N} \sum_{i=1}^{N} |\vec{r}_i(t) - \vec{r}_i(0)|^2 \right\rangle, \tag{S2}$$

where N is the number of molecules (of a given type) inside the pore, t is the simulation time, $\vec{r}(t)$ is the position vector at time t, and d is the dimension of the system. The only requirement for the application of the Einstein equation is that the atoms or molecules in the system move randomly due to elastic collisions, *i.e.* Brownian motion.

One readily recognises the term inside the angle brackets as the mean squared displacement (MSD) computed from the molecular trajectories $\vec{r}_i(t)$. Thus, the diffusion coefficient can be calculated from the slope of a linear fit of the $\mathrm{MSD}(t)$ curve according to

$$D_\mu = \frac{1}{2d} \frac{d}{dt} \mathrm{MSD}(t). \tag{S3}$$

The *normal* diffusion regime is defined as the region in which the MSD changes linearly with time. But depending on the nature of the framework-molecule interactions, other (non-linear) diffusion regimes may manifest in the $\mathrm{MSD}(t)$ curve.[74] These are often called *anomalous* diffusion regimes and are described by a power law, $\mathrm{MSD}(t) = K_\alpha t^\alpha$ where K_α is the "generalized diffusion coefficient". Several examples of different diffusion regimes are illustrated on Figure S3 and are listed below.

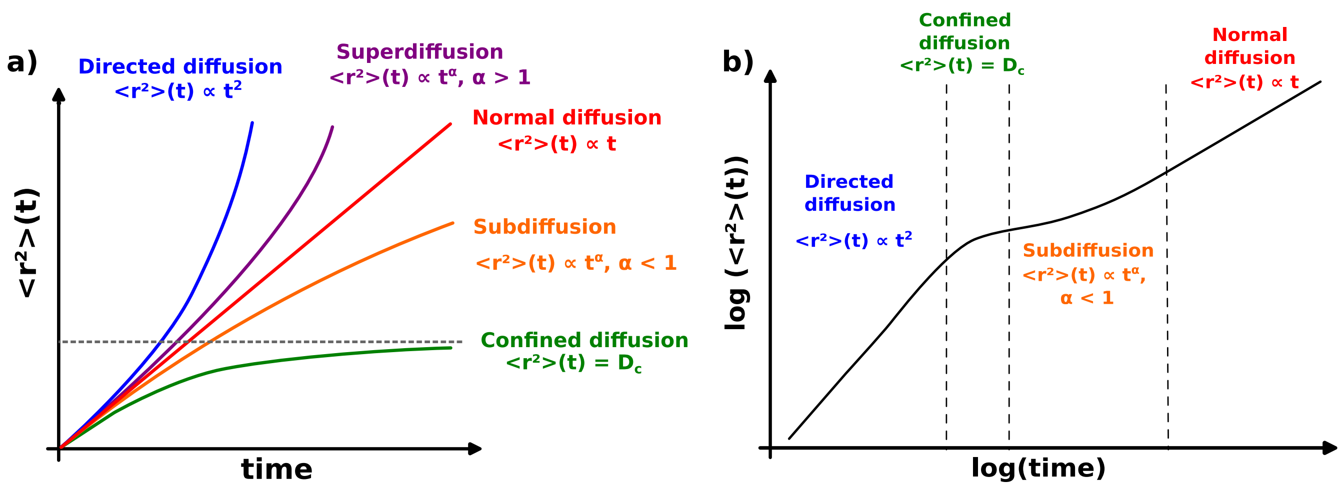


Figure S3. Different diffusion regimes and their dependence on α. (a) MSD curve for different diffusion regimes and (b) log-log plot of the MSD curve for a typical molecular diffusion simulation, illustrating different regimes that may occur.[74]

- **Directed diffusion** ($\alpha = 2$): ballistic motion, similar to a particle moving at constant speed.
- **Superdiffusion** ($2 > \alpha > 1$): complex interplay between accelerating forces and dispersive interactions between the particles and its neighbors.
- **Normal diffusion** ($\alpha = 1$): regular Brownian motion.
- **Subdiffusion** ($0 < \alpha < 1$): dominated by dispersive interactions between the particle and its neighbors.
- **Confined diffusion** ($\alpha = 0$): when particles are confined to cages with no longe-range diffusion.

Given that both Equation S2 and Equation S3 only apply to the normal regime, one must first locate the MSD region that obeys this regime in order to extract the diffusion coefficient. This procedure is usually performed by visual inspection combined with simple heuristics. Although this approach might work when simulating a few materials, it is not compatible with high-throughput studies.

Here, we propose the following approach for the automated extraction of self-diffusion coefficients from the MSD curve:

1. Check if MSD is greater than the largest axis of the simulation box (λ); If not, consider “confined diffusion”; Otherwise...

2. Locate inflection points of log-log MSD(t) curve by taking its second derivative;

3. Split MSD(t) curve into segments at the inflection points;

4. Identify diffusion regime in each segment by fitting a linear slope;

5. Locate normal diffusion regime as the region with the slope closest to 1;

6. Calculate self-diffusion coefficient using Equation S3 in the normal diffusion regime.

We illustrate in Figure S4 the decision-tree algorithm used to obtain the diffusion coefficient. Equation S3 is only applied to regions in which MSD $> \lambda$ and $\alpha \approx 1$. When a case of “confined diffusion” (MSD $< \lambda$ and $\alpha \approx 0$) is identified, the average of the MSD is taken as the confined diffusion coefficient D_c. Alternatively, when MSD $< \lambda$ but $\alpha \not\approx 0$, we conclude that the simulation does not contain enough time steps and needs to be run for longer. Finally, when MSD $> \lambda$ but $\alpha \not\approx 1$, we treat it as a case of anomalous diffusion and do not calculate any diffusion coefficient.

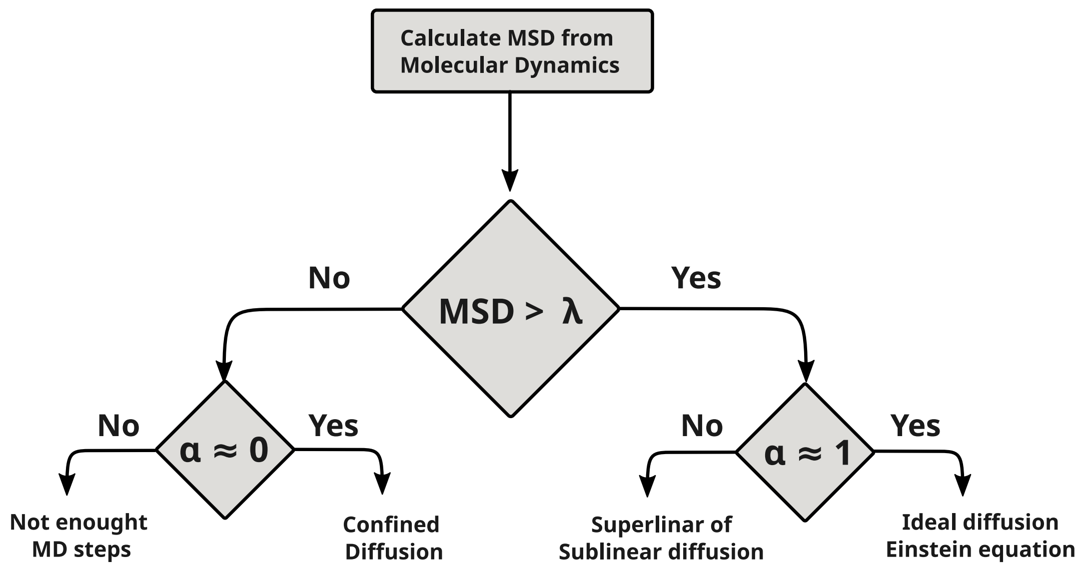


Figure S4. Flowchart representing the algorithm used to identify diffusion regimes in the MSD curve. This decision tree is applied to each section of the MSD curve delimited by its inflection points. Here, α corresponds to the fit coefficient and λ represents the maximum perpendicular distance of the simulation box.

S2.2.2 Distribution of diffusion coefficients for different adsorbates

We simulate the diffusion of CO_2 and N_2 molecules inside a host of selected materials. We plot in Figure S5a the distribution of CO_2 and N_2 heats of adsorption extracted from the dominant site of the Langmuir model that best fits the single-component isotherms. The symbols are coloured based on the balance between the microscopic diffusion coefficients: red for $D_\mu^{CO_2} \gg D_\mu^{N_2}$ and blue for $D_\mu^{CO_2} \ll D_\mu^{N_2}$. The vast majority of points lie below the parity line and, therefore, display higher adsorption affinity towards CO_2 (*i.e.* $-\Delta H_{CO_2} > -\Delta H_{N_2}$). On the other hand, the diffusion coefficient tends to decrease as the adsorption affinity increases. This negative correlation between adsorption and diffusion figures-of-merit generates competing effects at the process level and can hamper performance in materials with high CO_2/N_2 selectivity but low CO_2 diffusion.

In Figure S5b, the $D_\mu^{CO_2}$ *vs.* $D_\mu^{N_2}$ marginal plot shows a bimodal distribution with a preponderance of $D_\mu^{N_2} \gtrapprox D_\mu^{CO_2}$ points (slightly above the parity line). The median diffusion coefficient is 2×10^{-9} m^2/s for CO_2 and 9×10^{-9} m^2/s for N_2. Although the greater part of the distribution lies between 10^{-14} and 10^{-6} m^2/s, approximately 13% of the materials have diffusion coefficients smaller than 10^{-18} m^2/s for either CO_2 (bright blue cluster), N_2 (bright red cluster) or both (faint red/blue cluster on the lower left).

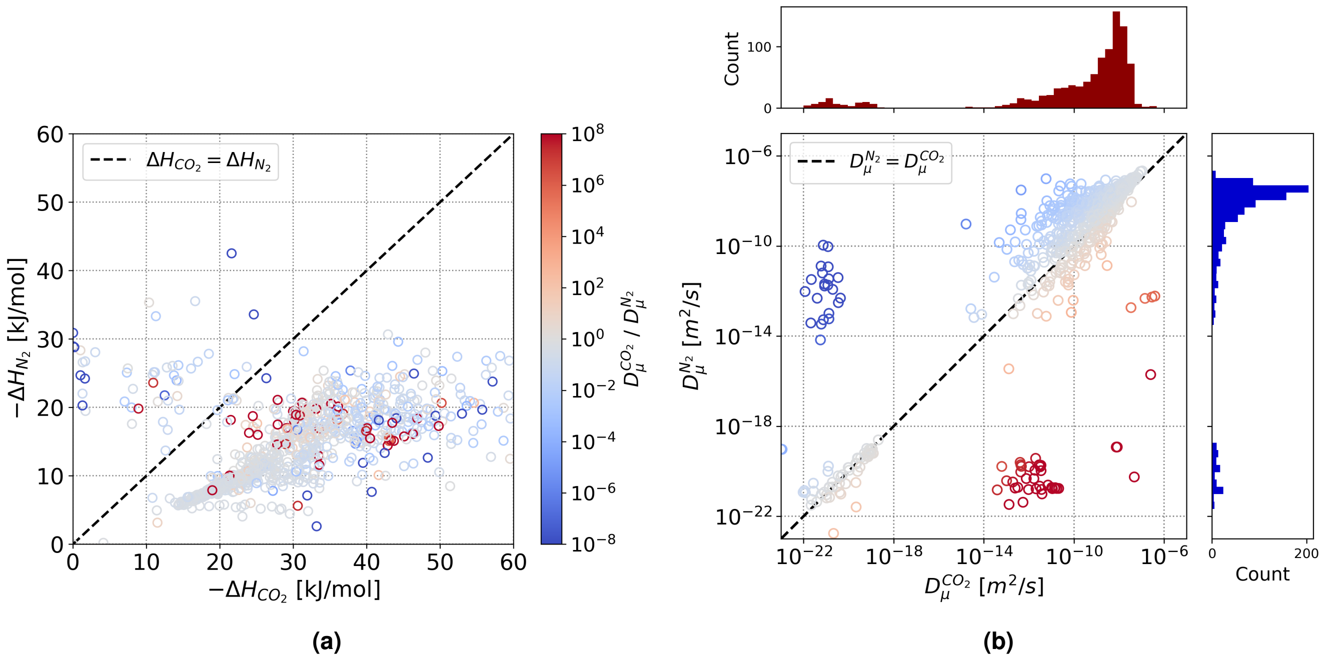


Figure S5. **Distribution of CO_2 and N_2 diffusion coefficients.** (a) Correlation between heat of adsorption and diffusion coefficient. Materials with high $\Delta H_{CO_2}/\Delta H_{N_2}$ ratio tend to display lower $D_\mu^{CO_2}/D_\mu^{N_2}$ ratio, which can hamper CO_2 capture productivity at process level. (b) Marginal plot of the bimodal distributions of diffusion coefficients for CO_2 and N_2, showing a slight preponderance of $D_\mu^{N_2} \gtrapprox D_\mu^{CO_2}$ values of the order of 10^{-9} m^2/s.

S2.2.3 Temperature-dependent diffusion coefficients

Ideally, the procedure above should have been applied to, not one, but *many* independent trajectories representing an average of MSD(t) curves in order to decrease the statistical uncertainty of the diffusion coefficient so-obtained. To alleviate this limitation, we perform molecular diffusion simulations at eight temperatures – from 348 to 698 K at regular intervals – and fit an Arrhenius-type equations to the data points, as shown in Figure S6. The temperature-dependent diffusion model can be expressed as

$$D_\mu(T) = D_0 \exp\left(\frac{-E_a}{k_B T}\right) \tag{S4}$$

where E_a denotes the energy barrier for the diffusion process, D_0 is the exponential pre-factor, k_B is the Boltzmann constant, and T the temperature. This equation can be linearised by taking the logarithm on both sides.

The fit was performed using `scipy` version 1.7.4,[75] and incorporating the uncertainty as relative weights for the fit. The bounds $-\infty < E_a < 0$ and $0 \leq D_0 \leq 1$ were imposed to ensure the physical consistency of the model. The use of the uncertainty as relative weights in the fit was important because it deemphasised outlier points gently, eliminating the need for data inspection and manual removal of outliers, which would introduce human bias and hamper reproducibility.

Materials with narrow pores, as well as materials with very high affinity for CO_2 due to open metal sites demonstrate thermally-activated diffusion. These materials display confined diffusion at lower temperatures and show a transition to normal diffusion at higher temperatures. To avoid such disparate diffusivity values, we intentionally remove any diffusivity below 10^{-19} m^2/s from the Arrhenius fit. Thus for some of the materials, the single component diffusivities are extrapolated to low temperature from their high temperature behavior.

S3 Process-based screening

We use a four-step temperature-swing adsorption (TSA) process model consisting of adsorption, purge, desorption and cooling steps, as depicted in Figure S7, to screen carbon capture materials. The dynamic TSA cyclic process model was developed using gPROMS® Model Builder software.

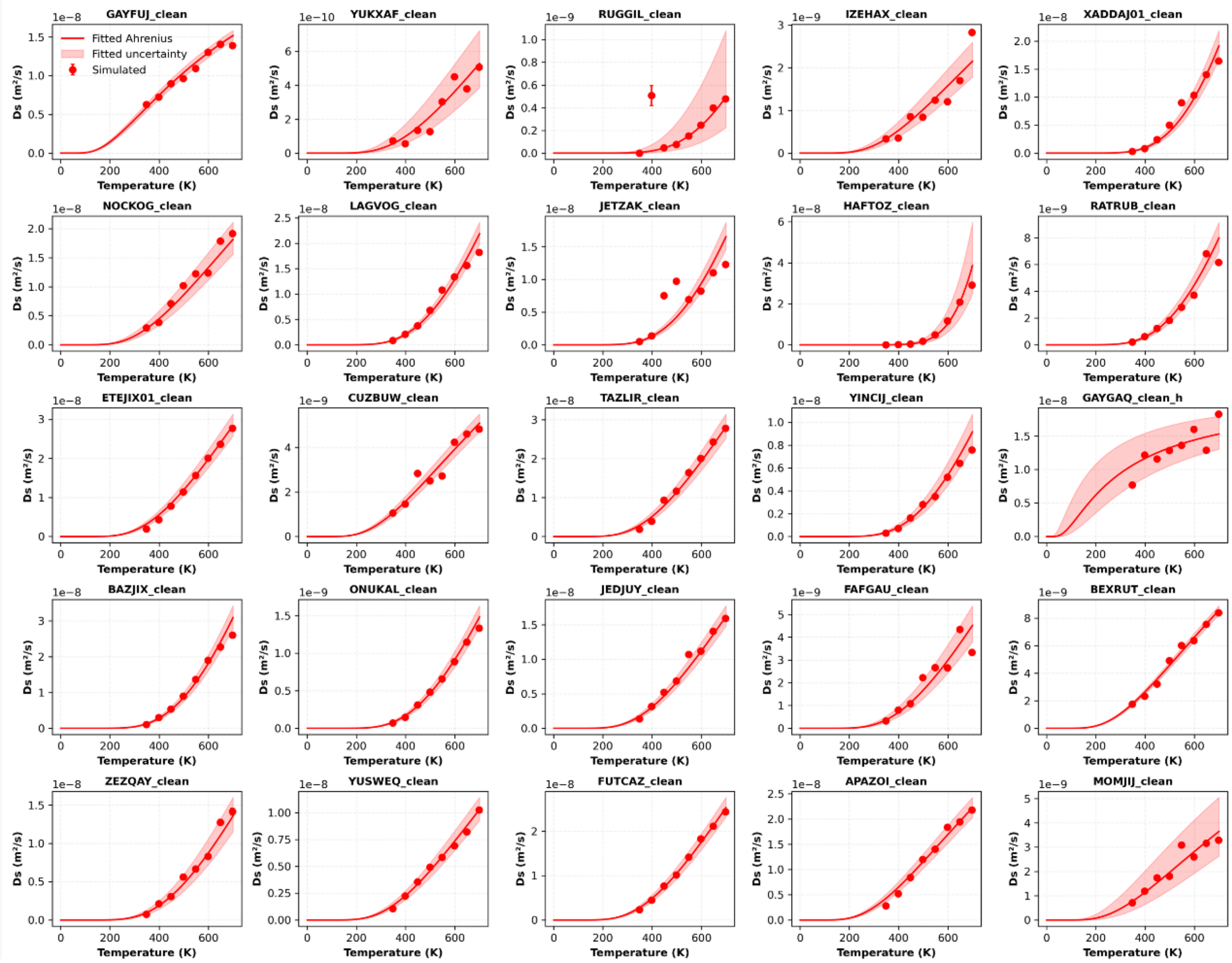


Figure S6. **Determination of temperature-dependent diffusion coefficients.** Examples of simulated diffusion coefficients (symbols) and adjusted Arrhenius expressions (lines) with associated fit uncertainty (shaded area).

S3.1 Temperature-swing adsorption process

The contactor for the process is assumed to be a fixed column with a packed adsorbent bed, as illustrated in Figure S7a. The heat transfer fluid flowing around the outer walls is always water, but at different temperatures, depending on the cycle step. Heating is achieved by simultaneously flowing steam and pressurized hot water, both at 105 °C, through the packed bed and heating jacket of the vessel, respectively. The system is cooled by pumping water at 30 °C through the heating jacket of the vessel, as well as purging the bed with room-temperature inert N_2. The temperature variation during the TSA cycle steps is shown in Figure S7b.

The cycle steps detailed in Figure S7c are described as follows. In the adsorption step, a flue gas stream consisting of 4.1% CO_2 and 95.9% N_2 enters the sorbent bed while cooling water flows through the outer jacket of the adsorption bed. We assumed dry flue gas as feed to simplify the process model and interpretation, with an implicit (not shown) moisture removal step before CO_2 capture. In this step, (mainly) CO_2 is adsorbed while the CO_2-depleted stream of N_2 flows out. In the subsequent purge step, the bed is heated as previously described, leading to N_2 desorption and purge from inter-particle bed volume. The N_2-rich purged gas is discarded. In the following desorption step, due to continued heating up to 105 °C, stronger bound CO_2 is desorbed and collected from the outlet as product alongside the steam used for heating the bed. This product stream is cooled to remove the steam/water before compressing CO_2 to required pressure for sequestrations. Both of these steps are not shown or accounted for in the process based screening. In the cooling step, the bed is brought down to 35 °C as previously described. During cooling, the bed is dried to remove H_2O using N_2 as inert stream. Since the cooling step is relatively small, the N_2 required for this step can be sourced from effluent of the adsorption step. Finally, the dry and cool bed undergoes cyclic TSA process again until cyclic steady state is achieved, based on having less than 0.1% cycle-to-cycle variation in the CO_2 purity and recovery.

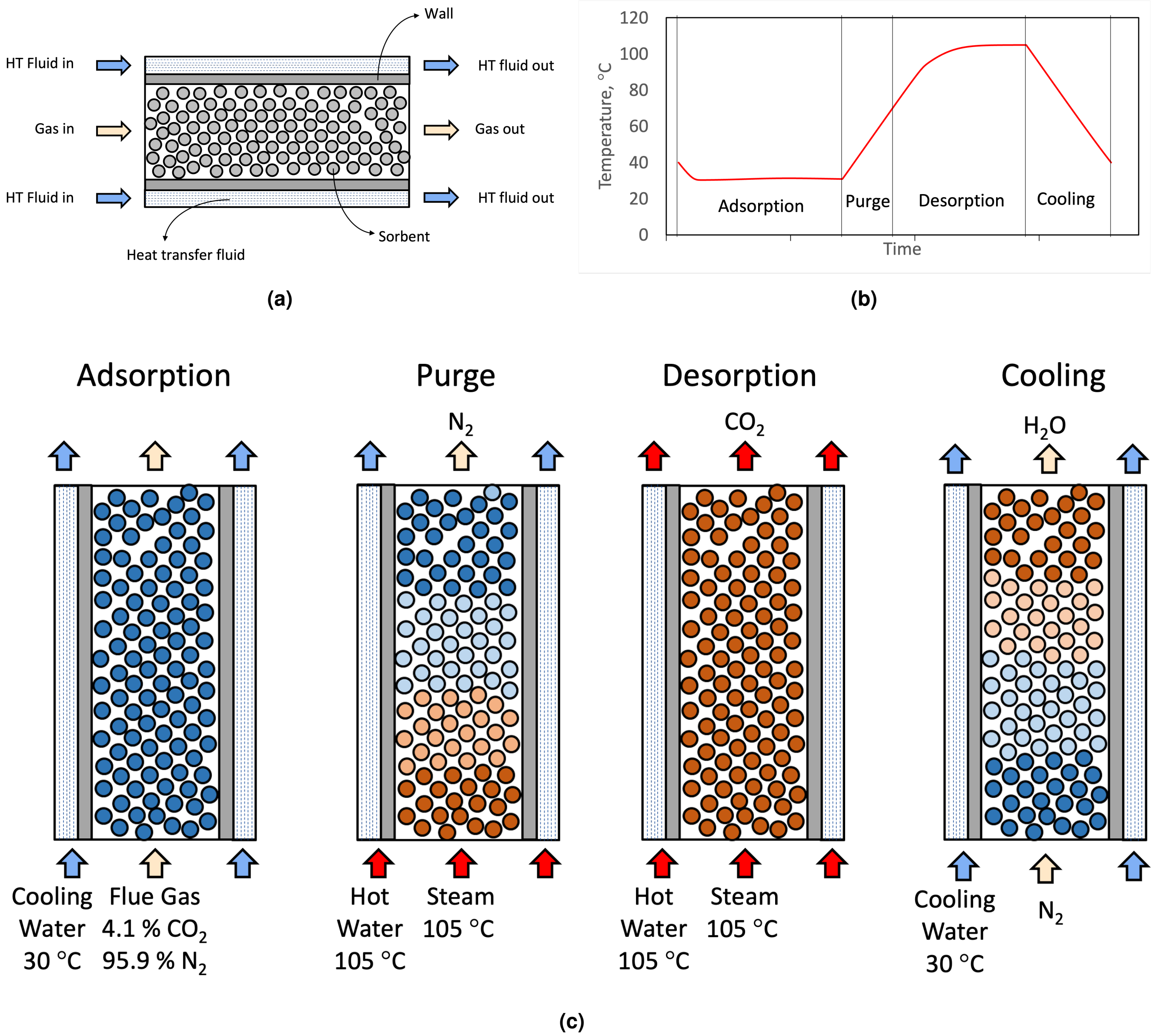


Figure S7. Temperature-swing adsorption (TSA) process model. (a) Illustration of the sorbent bed in contact with the flue gas. (b) Variation of bed temperatures during TSA cycle. (c) TSA cycle steps, including adsorption, purge, desorption and cooling. Purge and desorption steps use steam as the energy source. All steps use water, albeit at different temperatures, as heat-transfer fluid.

Mass balance is simulated for the various phases according to Equation S5. The meaning and values of the variables and parameters are described in Table S6. The effective linear driving mass transfer coefficient (K_{eff}) is described as Equation 2. This equation ignores pellet film resistance as well as does simplifying assumptions for pellet void fraction (ε_p) and thermodynamic correction (q^*/c) to enable consistent comparison between constant diffusivity and mass transfer with consideration for molecular diffusion.

$$
\begin{aligned}
\frac{\partial c_i}{\partial t} &= D_{\mathrm{ax}}\frac{\partial^2 c_i}{\partial z^2} - \frac{\partial\left(u_g c_i\right)}{\partial z} - \frac{(1-\varepsilon)\rho_s}{\varepsilon}\frac{\partial q_i}{\partial t} \\
\frac{\partial q_i}{\partial t} &= \sum_{j=1}^{S}\frac{\partial q_{i,j}}{\partial t} \\
\frac{\partial q_{i,j}}{\partial t} &= K_{\mathrm{eff}}\left(q_{\mathrm{eq},i,j} - q_{i,j}\right) \\
D_p^* &= D_p\frac{\varepsilon_p}{\rho_p}\frac{c}{q^*} \\
q_{\mathrm{eq},i,j} &= \frac{q_{\mathrm{sat},i,j}\, b_{i,j}\cdot c_{g,i}}{1+\sum_k b_{j,k}\cdot c_{g,k}} \\
b_{i,j} &= \exp\left(-\frac{\Delta H_{\mathrm{ads},i,j} - T\Delta S_{\mathrm{ads},i,j}}{RT}\right)
\end{aligned}
\tag{S5}
$$

The energy balance equations for the various phases and components are described by Equation S6 using the variables and parameters defined in Table S6. The properties of the gas phase (subscript g), sorbent phase (subscript s), wall of the contactor shell (subscript w) and the external heat transfer fluid (subscript e) are modelled as follows.

$$
\begin{aligned}
\frac{\partial T_g}{\partial t} &= \frac{1}{\varepsilon\rho_g C_g}\left[\lambda_g\frac{\partial^2 T_g}{\partial z^2} - \varepsilon\rho_g C_g u_g\frac{\partial T_g}{\partial z} + \frac{6(1-\varepsilon)}{2R_p}h_{g,s}\left(T_s - T_g\right) + \frac{\pi d_{is} h\varepsilon}{V_s}h_{g,w}\left(T_w - T_g\right) + \frac{\pi d_{ot} h_{nT}\varepsilon}{V_s}h_{g,w}t\right] \\
\frac{\partial T_s}{\partial t} &= \frac{1}{(1-\varepsilon)\rho_s C_s}\left[\lambda_s\frac{\partial^2 T_s}{\partial z^2} - (1-\varepsilon)^2\rho_s^2 C_s\sum_{i,k}\frac{\partial q_{i,k}}{\partial t}\Delta H_{i,k} + \frac{6(1-\varepsilon)}{2R_p}h_{g,s}\left(T_g - T_s\right) + \frac{\pi d_{is} h(1-\varepsilon)}{V_w}h_{s,w}\left(T_w - T_s\right)\right] \\
\frac{\partial T_w}{\partial t} &= \frac{1}{\rho_w C_w}\left[\lambda_w\frac{\partial^2 T_w}{\partial z^2} + \frac{\pi d_{is} h\varepsilon}{V_w}h_{g,w}\left(T_g - T_w\right) + \frac{\pi d_{is} h(1-\varepsilon)}{V_w}h_{s,w}\left(T_s - T_w\right) + \frac{\pi d_{os} h}{V_w}h_{e,w}\left(T_e - T_w\right)\right] \\
\frac{\partial T_e}{\partial t} &= \frac{1}{\rho_e C_e}\left[\lambda_e\frac{\partial^2 T_e}{\partial z^2} - \rho_e C_e u_e\frac{\partial T_e}{\partial z} + \frac{\pi d_{os} h}{V_e}h_{w,e}\left(T_w - T_e\right)\right]
\end{aligned}
\tag{S6}
$$

Momentum balance for the column is solved using Ergun equation as shown in Equation S7 where velocity and viscosity of gas are changing along the axis of the column. The meaning and values of the variables and parameters are described in Table S6.

$$
-\frac{\partial P}{\partial z} = 150\frac{\mu_g}{\left(2R_p\right)^2}\frac{(1-\varepsilon)^2}{\varepsilon^3}\left(\varepsilon u_g\right) + 1.75\frac{\rho_g}{2R_p}\frac{(1-\varepsilon)}{\varepsilon^3}\left(\varepsilon u_g\right)^2 \tag{S7}
$$

S3.2 Process-level performance metrics

To illustrate the output of the process model we have selected 4 example MOFs, namely `VASFON`, `BETGUE`, `MOCHIW` and `EVOMOR`. These MOFs have been selected to show the impact of the adsorption capacity (isotherm) and diffusion coefficient on the performance and rank order of the MOFs. Out of these 4 selected MOFs, `VASFON` and `MOCHIW` are higher performing (Higher Ranked) MOFs and `BETGUE` and `EVOMOR` are lower performing (lower rank order) MOFs. `VASFON` and `BETGUE` show lower sensitivity towards crystal size while `MOCHIW` and `EVOMOR` show higher sensitivity towards crystal size due to diffusion limitation.

We show structures and isotherm information for these 4 MOFs inFigure S8. The isotherm plots include the simulated CO_2 and N_2 adsorption capacities for these MOFs at multiple pressures and 4 different temperatures. The fitted isotherm model is shown on the same plot. Goodness of fit (Root Mean Square Error - RMSE) is used to The higher performing MOFs `VASFON` and `MOCHIW`, have significant capacity at 0.041 bar of CO_2 partial pressure while MOFs `BETGUE` and `EVOMOR` have lower CO_2 adsorption capacity at lower partial pressures.

In Figure S9, we show diffusion constants from the molecular dynamic simulations performed for above mentioned MOFs and fitted activated diffusion model by lines. `VASFON` and `BETGUE` show higher diffusivity for both CO2 and N2 - with diffusion constant between 10^{-9} to 10^{-8} m^2/s depending on temperature. On the other hand `MOCHIW` and `EVOMOR` show limited

[a]Ideal gas law.
[b]Based on ideal crystal density.
[c]Machine-learning model from Moosavi *et al.*[76]

Table S6. Process Model Parameters: Notations and values

Symbol	Description	Value(s)
	Bed Geometry	
	Bed Length	1 m
d_{is}	Packed Bed inside Diameter	3.00 cm
d_{os}	Packed Bed outside Diameter	3.18 cm
d_{ot}	Heat Exchange Jacket inside Diameter	3.36 cm
$V_s \mid V_w \mid V_e$	Total volume of {sorbent bed, wall, HTF}	Based on geometry
ε	Bed void fraction	0.35
	Conditions and TSA Cycle Details	
	Bed Outlet Pressure	1.0 bar
	Cycle Step Time (Ads | Purge | Des | Cool)	Determined by Optimizer
	Feed Gas Type (Ads | Purge | Des | Cool)	Flue | Steam | Steam | Nitrogen
	Feed Gas Flow Rate (Ads | Purge | Des | Cool)	200 | 60 | 60 | 200 cm^3/s
	Feed Gas Temperature (Ads | Purge | Des | Cool)	30 | 105 | 105 | 30 °C
	Heat Exchange Fluid Temperature (Ads | Purge | Des | Cool)	30 | 105 | 105 | 30 °C
$\rho_g \mid \rho_s \mid \rho_w \mid \rho_e$	Density of the {gas | sorbent | wall | HTF}	{[a] | [b] | 8 | 0.866} g/cm^3
$T_g \mid T_s \mid T_w \mid T_e$	Temperature of the {gas | sorbent | wall | HTF}	Output of Equation S6
u_e	Inlet Exchange Fluid Velocity	0.2 m/s
u_g	Gas interstitial velocity	
P	Local pressure in sorbent bed	Output of Equation S7
μ_g	Viscosity of gas	1.9^{-5} Pa.s
	Sorbent Details and Mass Transfer	
R_p	Sorbent particle radius	2 mm
R_μ	Sorbent crystal radius	0.5, 5, 50 µm
D_p	Sorbent intra-particle diffusivity	
D_p^*	Sorbent intra-particle effective diffusivity	5×10^{-8} m^2/s
D_μ	Sorbent intra-crystal diffusivity	
K_{eff}	Effective mass transfer coefficient	Output of Equation 2
D_{ax}	Axial dispersion coefficient	5×10^{-7} m^2/s
	Heat Transfer	
$\lambda_g \mid \lambda_s \mid \lambda_w \mid \lambda_e$	Thermal conductivity of the {gas | sorbent | wall | HTF}	{0.03 | 0.5 | 15 | 0.13} W/(m.K)
$h_{g,s} \mid h_{g,w} \mid h_{s,w} \mid h_{e,w}$	Heat transfer coef {gas-sorbent | gas-wall | sorbent-wall | HTF-wall}	200 W/(m^2.K)
$C_g \mid C_s \mid C_w \mid C_e$	Heat Capacity of the {gas | sorbent | wall | HTF}	{1000 | [c] | 466 | 3075} J/kg/K
ρ_w	Density of the contactor's wall	8 g/cm^3
ρ_e	Density of the external heat transfer fluid	0.866 g/cm^3
	Adsorption Isotherm	
q_i	Concentration of component i in the adsorbed phase	Output of Equation S5
c_i	Concentration of component i in the gas phase	Output of Equation S5
$q_{\mathrm{eq},i}$	Equilibrium concentration of component i in the adsorbed phase	Output of Equation S1
$b_{i,j}$	Langmuir isotherm based adsorption affinity	Output of Equation S5
$\Delta H_{\mathrm{ads},i,j}$	Fitted enthalpy of adsorption for site i and species j	non-linear optimization
$\Delta S_{\mathrm{ads},i,j}$	Fitted enttropy of adsorption for site i and species j	non-linear optimization
$i \mid k$	Index that spans adsorbates	$i = \{CO_2, N_2, H_2O\}$
j	Index that spans Langmuir sites	$0 < j <= S$
S	Number of Langmuir sites considered	$S = \{1, 2, 3\}$

diffusivity for both CO_2 and N_2. As shown in Table ?, both `MOCHIW` and `EVOMOR` show limited pore limiting diameter (PLD). For `MOCHIW`, the diffusivities based on higher temperature simulations, are 10^{-14} to 10^{-12} m^2/s. For `EVOMOR` diffusivities are nearly very low (10^{-20} to 10^{-19} m^2/s.

In Figure S10, we show cyclic simulation output in terms of the mole fraction of gas exiting the TSA column for CO_2, N_2

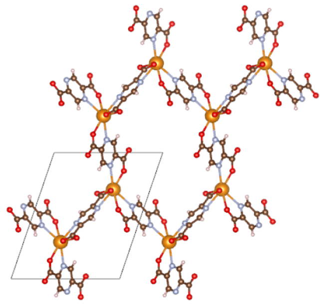
and H_2O. We also show column temperature at the exit of the adsorption column as a function of cycle time. The cycle time are shown by vertical lines on the cyclic profile. For `VASFON` and `BETGUE` the cyclic profile and process based performance of the CO_2 separation is same irrespective of the crystal size with molecular diffusion or with constant pellet diffusion. Hence, we report only cyclic simulation profile for constant diffusivity. Comparing Figure S10a and Figure S10b, `VASFON` with higher capacity has longer adsorption step while `BETGUE` with lower CO_2 capacity has shorter adsorption step. For `MOCHIW` and `EVOMOR` due to lower microscopic diffusivity, show cyclic simulation profile changing as a function of crystal size showing separation deteriorating with increasing crystal size. Comparing constant diffusivity cyclic profiles of `MOCHIW` (Figure S10c and `EVOMOR` (Figure S10e), they show similar behavior as of `VASFON` and `BETGUE`. However, comparing cyclic profiles for constant diffusivity (Figure S10c) and variable diffusivity (Figure S10d) for `MOCHIW`, shows the cycle becoming shorter with earlier CO_2 breakthough reducing the separation performance with inclusion of the microscopic diffusivity. Similarly poorly performing MOF `EVOMOR`, cycle becomes even shorter with the inclusion of the diffusivity. The short cycle due to very low diffusivity of CO_2 makes concentration and temperature arbitrary from the process optimization. For `EVOMOR` with diffusivity considerations (Figure S10f, the CO_2 breakthough is observed very early in shortest adsorption step along with N_2 leading to no selectivity due to very low diffusivity, thereby limiting the possible purity or recovery in cyclic separation process.

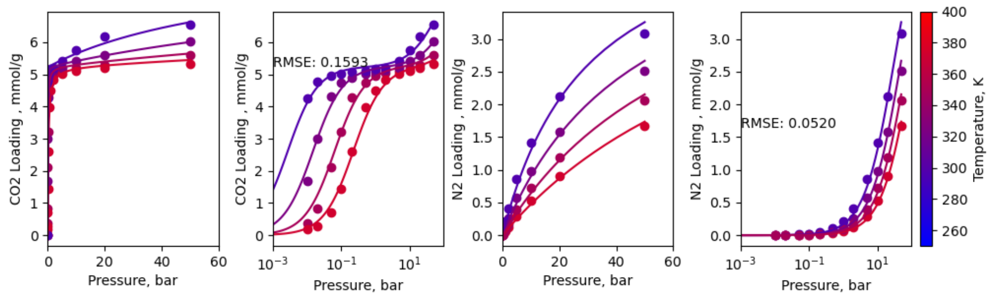
The process modeling outcome also includes the tradeoffs between the process based figures of merits (CO_2 purity, recovery, specific energy consumption and volumetric productivity) reported in Figure S11. As shown in Figure S11a and Figure S11b, the purity vs. recovery and specific energy vs productivity are unchanged as a function of diffusivity. In ca case of the `MOCHIW` Figure S11c, the purity vs recovery pareto fronts show gradual shift from top right corner to bottom left, moving from 1 μm crystal size to 10 μm and 100 μm. The maximum recovery possible decreases from near 98% for 1 μm crystal to 89% for 10 μm to 18% for 100 μm crystal. For `EVOMOR`, due to very poor diffusivities, the separation performance of MOF goes from good to very poor by just changing constant diffusivity to any crystal size as shown in Figure S11d

A representative MOF ranking is given in Table S7 for the 20 highest-ranked MOFs by volumetric productivity at constant K_{eff}. For the same set of MOFs, the rank-order by volumetric productivity is also provided for crystallite diameters R_μ =1 μm, 10 μm, and 100 μm, respectively. These rankings account for structure-dependent, molecular diffusivity. As examples of the influence of diffusivity and crystallite diameter on rank-order, `VASFON` maintains second place in the ranking and becomes the highest-ranked MOF at R_μ =100 μm while (`BOHGOU`) drops out of the TOP-20 list. The MOF structure `MOCHIW` ranks successively lower as R_μ increases, falling from rank #12 to #495 at R_μ =100 μm.

Table S7. TOP-20 MOF ranked by volumetric productivity at a 95% CO_2-purity level for various crystallite diameters R_μ

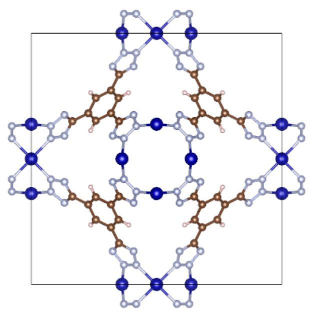
MOF Name	K_{eff}=const.	R_μ =1 μm	R_μ =10 μm	R_μ =100 μm
BOHGOU	1	1	1	177
VASFON	2	2	2	1
NATXIR	3	3	3	2
CDLGLU11	4	698	695	721
VASFED	5	4	4	4
CAKXET	6	5	5	19
AZAFIS	7	6	6	3
MOCJAQ	8	7		
ESUMOU	9	697	692	675
CDLGLU01	10			
MUXHIW01	11	8	59	
MOCHES	12	11	88	495
IKETAU	13	9	7	9
RATVEP	14	10	9	21
MOCHIW	15	17	138	485
SUKREX	16	12	8	5
IPIJUM	17	13	10	6
NARMUR	18	14	11	12
MOCJEU	19	15	18	
VIDXOX	20	16	12	7

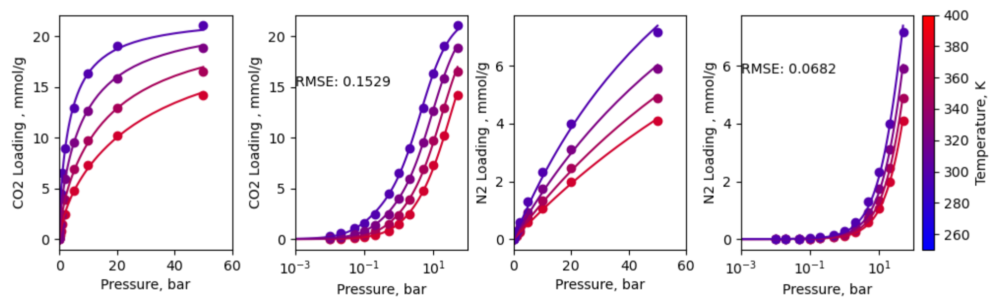




(a) VASFON

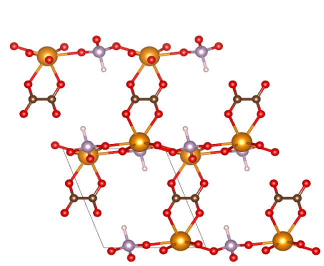
(b)

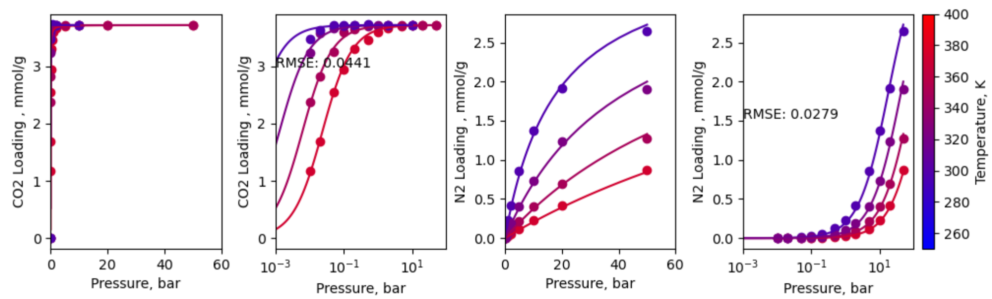




(c) BETGUE

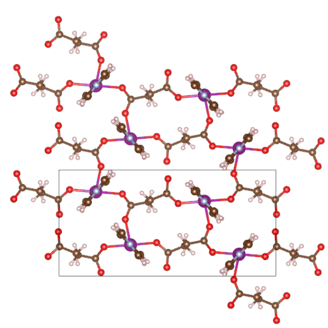
(d)

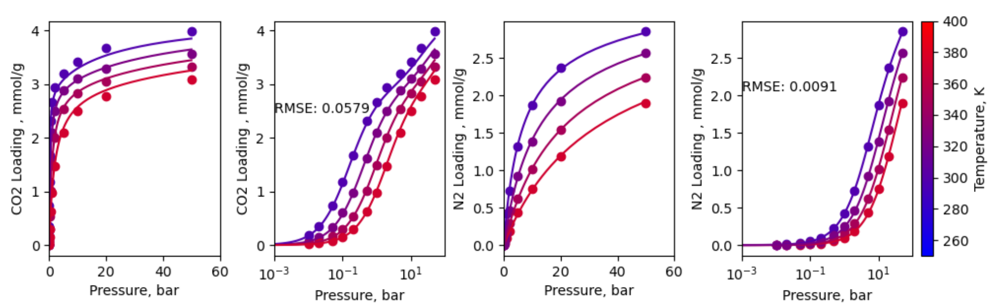




(e) MOCHIW

(f)





(g) EVOMOR

(h)

Figure S8. MOF Structures and Isotherms (a) `VASFON`, (c) `BETGUE`, (e) `MOCHIW` and (g) `EVOMOR` structures and their associated isotherms (b), (d), (f) and (h), respectively. CO_2 and N_2 isotherm data (filled circles) and fitted models (lines).

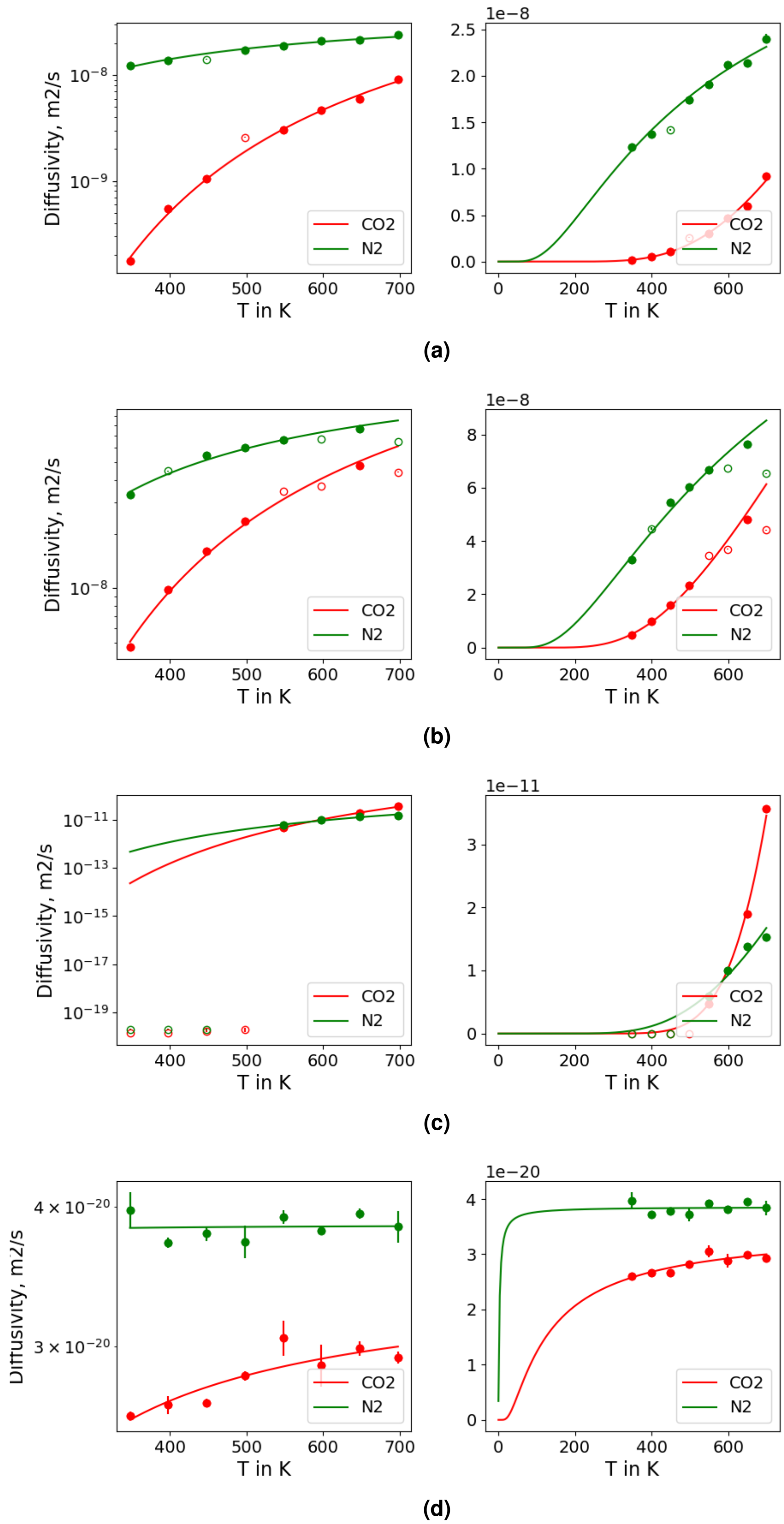


Figure S9. **Diffusion Coefficients** CO_2, N_2 diffusivity data (filled and empty circles) and fitted models (lines) using data represented by filled circle for MOFs (a) `VASFON` (b) `BETGUE`, (c) `MOCHIW`, (d) `EVOMOR` in both normal and semi-log scale.

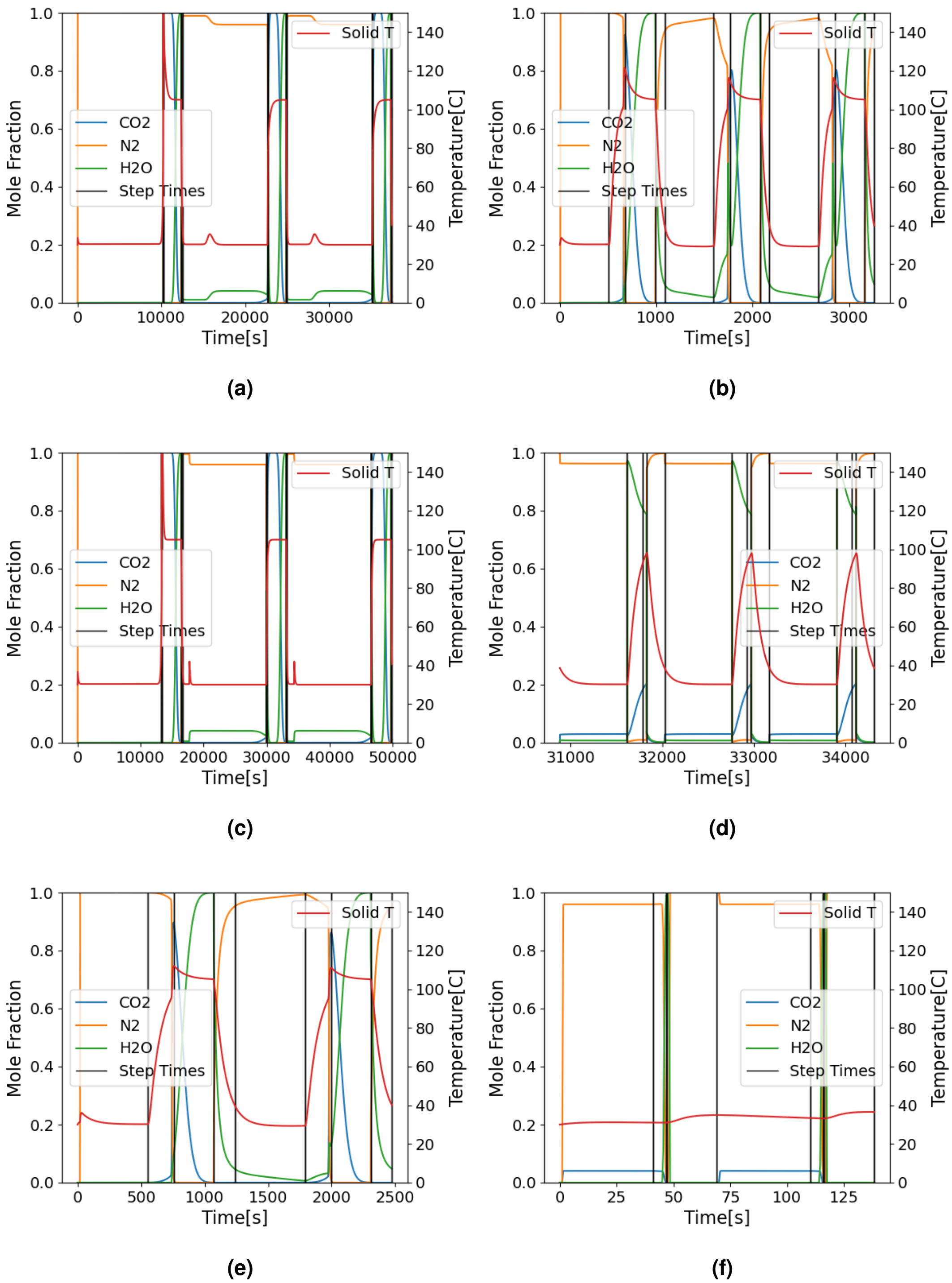
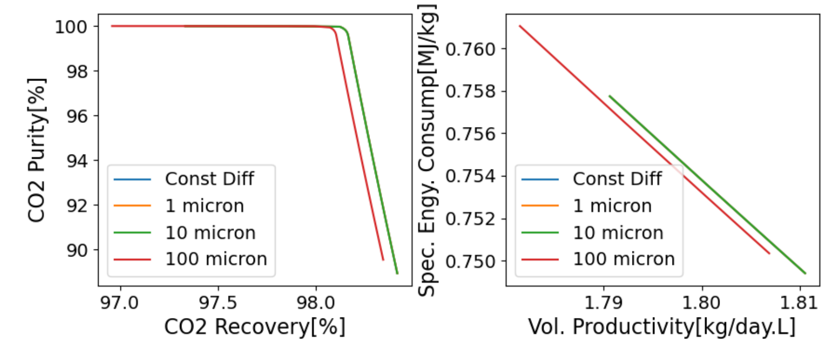
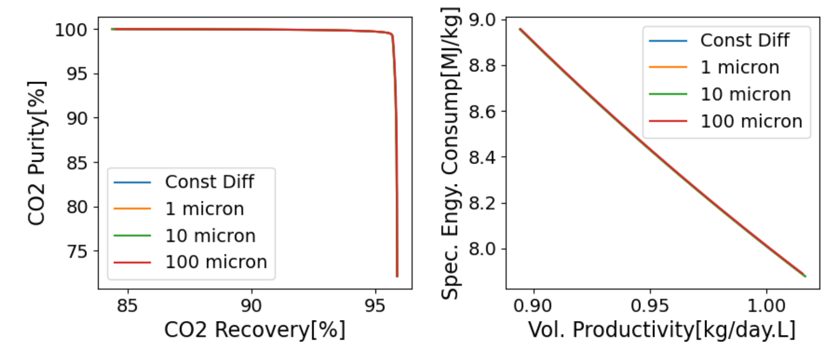


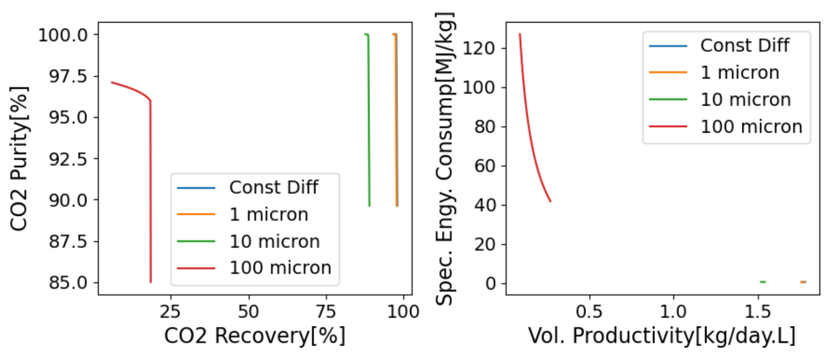
Figure S10. Process Model TSA Model Output CO_2, N_2 and H_2O mole fractions and adsorbent temperature at the exit of the column plotted as function of time for (a) `VASFON` with constant diffucivity (b) `BETGUE` with constant diffusivity, (c) `MOCHIW` with constant diffusivity, (d) `MOCHIW` with variable diffusivity for 100 μm, (e) `EVOMOR` with constant diffusivity, (e) `EVOMOR` with variable diffusivity for 100 μm.



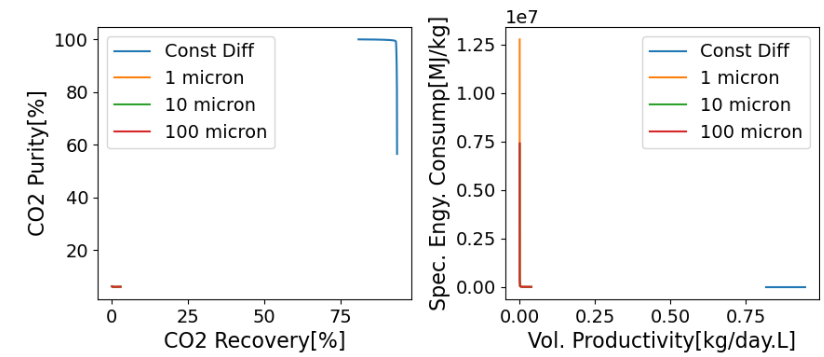
(a)



(b)

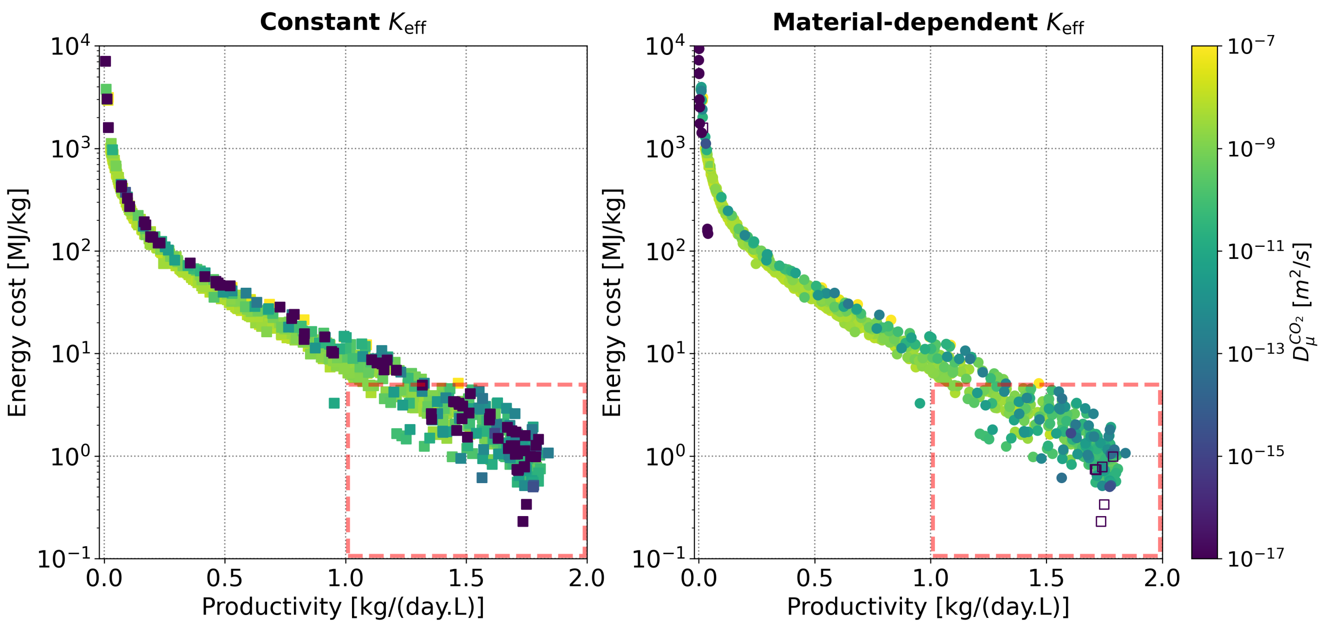


(c)

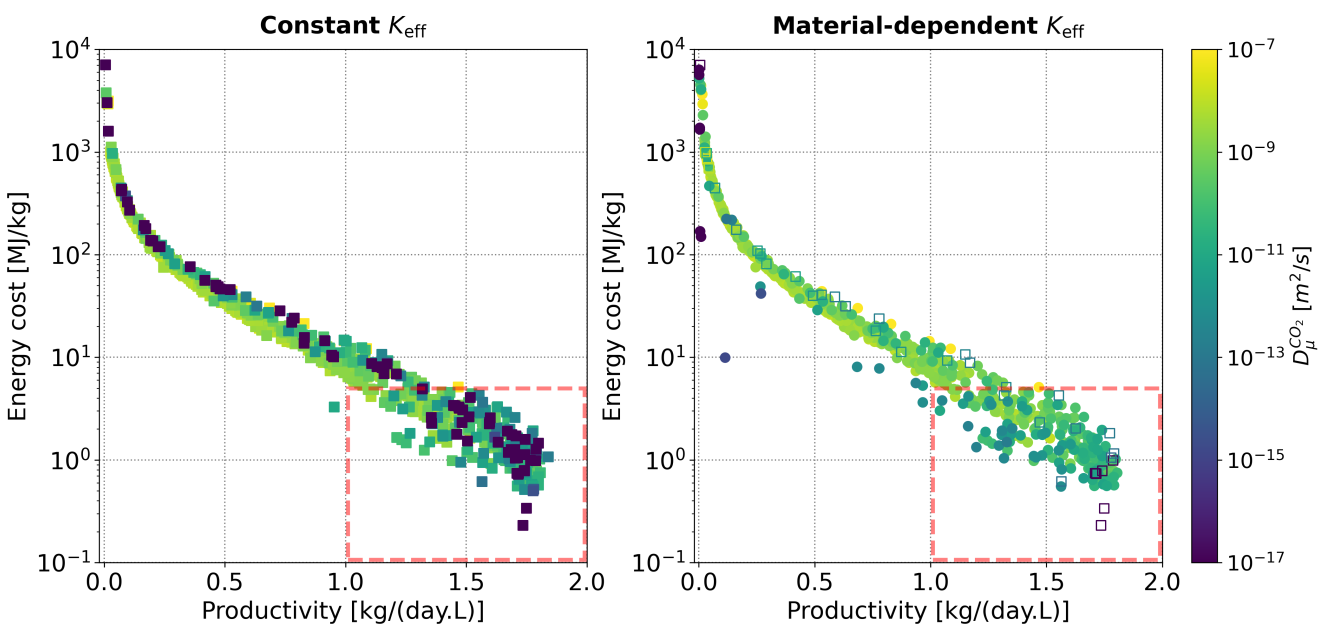


(d)

Figure S11. **Pareto Plots for Process Based Figures of Merits** Purity [%] vs. Recovery [%] on left and Specific Energy [MJ/kg of CO_2] vs. volumetric productivity [kg of CO_2/day/L of sorbent] on right, for MOFs (a) `VASFON` (b) `BETGUE`, (c) `MOCHIW`, (d) `EVOMOR`.



(a)



(b)

Figure S12. Effect of crystallite size on energy–productivity metrics. Distribution of energy cost *vs.* productivity (at 95% purity) metrics for crystallite sizes (a) $R_\mu = 1$ μm and (b) $R_\mu = 100$ μm, assuming the same constant K_{eff} for all materials (left) and with material-dependent K_{eff} (right). Open symbols in the right panel represent materials for which the process-level optimisation did not yield 95% purity, so the coordinates of the left panel were used instead. Line colours in the right panel represent the CO_2 diffusion coefficient.